

EIDETIC POWER-MAP ALGEBRAS: BALANCED KERNELS, RATIONAL CLASSES, AND RAMIFIED COMPLETION

SIMON VELEZ

ABSTRACT. For a finite group G , let V_G be the rational class functions with the inner product weighted by class sizes, and let Ψ_n be the power map $f(C) \mapsto f(C^n)$. The ramified power-map algebra $\mathcal{P}_{\text{ram}}(G)$ is the rational algebra generated by Ψ_p and its adjoint Ψ_p^* for the primes p dividing $\exp(G)$. It is an invariant of the class table enriched by power maps, and it detects information the ordinary character table does not: D_8 and Q_8 have dimensions 17 and 10.

We prove that $\mathcal{P}_{\text{ram}}(G)$ is semisimple and equal to its own bicommutant, identify the commutant with the balanced kernels on the weighted power graph, and show that the transfer identity $\Psi_p^* \Psi_p = M_{r_p}$ makes the algebra full exactly when the only balanced kernels are scalar. The power maps prime to the exponent act as Frobenius elements: they commute with $\mathcal{P}_{\text{ram}}(G)$, they identify V_G with the Galois representation of the étale algebra of character-value fields, and their determinants give the complete Euler product of its Dedekind zeta function, with the ramified factors recovered from inertia invariants. When the commutant of $\mathcal{P}_{\text{ram}}(G)$ is commutative, every simple factor carries a Dirichlet character of the unit group, trivial or quadratic when the center is $\mathbb{Q}$, and $\mathcal{P}_{\text{ram}}(G)$ coincides with the algebra generated by all power maps.

We compute the algebra for groups of prime exponent, cyclic prime-power groups, dihedral groups of order $2p$, and all twenty-six sporadic simple groups. Every sporadic factor has Schur index one; the centers are $\mathbb{Q}$, nine copies of $\mathbb{Q}(\zeta_3)$, and one $\mathbb{Q}(\zeta_5)$, and the nonrational centers occur exactly for the six pariah groups. For the Monster, a 186-vector certificate proves

$$\mathcal{P}_{\text{ram}}(\mathbb{M}) \cong M_{170}(\mathbb{Q}) \oplus M_5(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q})^{\oplus 4} \oplus \mathbb{Q}^{\oplus 8},$$

the centralizer of a symmetry group $(C_2)^{14}$ of the weighted power system. One generator of that group swaps $16B$ with $16C$ and $32A$ with $32B$; it preserves all class sizes and power maps but is not an automorphism of the character table.

1. INTRODUCTION

Let G be a finite group, $\text{Conj}(G)$ its set of conjugacy classes ordered so that the identity class is first, and $V_G = \text{Map}(\text{Conj}(G), \mathbb{Q})$ the space of $\mathbb{Q}$ -valued class functions. For $n \geq 1$ the power map $[x] \mapsto [x^n]$ induces an operator

$$(\Psi_n f)(C) = f(C^n),$$

where C^n is the class of n th powers of the elements of C . The normalized class form

$$\langle f, h \rangle_G = \frac{1}{|G|} \sum_C |C| f(C) h(C)$$

is positive definite, and its adjoint Ψ_n^* records the weighted reverse fibres of the power map.

The construction depends only on the conjugacy classes, their sizes, and the selected power maps. It is therefore an invariant of the power-map-enriched class table.

For a label set $E \subseteq \mathbb{N}_{>0}$ containing 1, write

$$\mathcal{P}_E(G) = \text{Alg}_{\mathbb{Q}}\{I, \Psi_n, \Psi_n^* : n \in E\} \subseteq \text{End}_{\mathbb{Q}}(V_G) \quad (1.1)$$

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for the rational adjoint-closed algebra generated by these operators. The E -marked eidetic object is

$$\mathbf{E}_E(G) = (V_G, \mathcal{B}_G, \langle \cdot, \cdot \rangle_G, \{\Psi_n\}_{n \in E}, \mathcal{P}_E(G)), \quad (1.2)$$

where $\mathcal{B}_G$ is the class-coordinate basis. The canonical label set is the set of primes dividing the exponent, $E = \pi(\exp(G))$; then $\mathcal{P}_{\text{ram}}(G) = \mathcal{P}_E(G)$ is the *ramified power-map algebra*. The *ramified eidetic object* is

$$\mathbf{E}_{\text{ram}}(G) = (V_G, \mathcal{B}_G, \langle \cdot, \cdot \rangle_G, \{\Psi_n\}_{n \geq 1}, \mathcal{P}_{\text{ram}}(G)). \quad (1.3)$$

The specialization $E = \mathbb{N}_{>0}$ gives the *full Adams algebra* $\mathcal{P}_{\text{Ad}}(G)$ and its associated object $\mathbf{E}_{\text{Ad}}(G)$.

1.1. What the algebra sees. The algebra uses class sizes and power maps, so it need not factor through the ordinary complex character table. It does not. The smallest classical example already gives a sharp separation.

Theorem 1.1. *The dihedral group D_8 and quaternion group Q_8 are isoclinic and have equivalent ordinary complex character tables. Nevertheless*

$$\dim_{\mathbb{Q}} \mathcal{P}_{\text{ram}}(D_8) = 17, \quad \dim_{\mathbb{Q}} \mathcal{P}_{\text{ram}}(Q_8) = 10,$$

so their represented rational power-map algebras are nonisomorphic. Over $\mathbb{C}$,

$$\mathcal{P}_{\text{ram}}(D_8) \otimes_{\mathbb{Q}} \mathbb{C} \cong M_4(\mathbb{C}) \oplus \mathbb{C}, \quad \mathcal{P}_{\text{ram}}(Q_8) \otimes_{\mathbb{Q}} \mathbb{C} \cong M_3(\mathbb{C}) \oplus \mathbb{C},$$

and the scalar Q_8 -constituent has multiplicity two in class-coordinate space.

Proof. Both groups have five ordinary conjugacy classes of sizes 1, 1, 2, 2, 2. After a permutation of the final three columns, their common ordinary character table is

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 \\ 2 & -2 & 0 & 0 & 0 \end{pmatrix}.$$

Let ρ be the two-dimensional irreducible representation. Then

$$\lambda^2(\rho) = \det(\rho) \neq 1 \quad \text{for } D_8, \quad \lambda^2(\rho) = 1 \quad \text{for } Q_8.$$

The standard representation of D_8 contains reflections and has nontrivial determinant, while $Q_8 \subseteq \text{SU}(2)$. Hence the ordinary-table identification does not preserve the based augmented λ -ring.

Their power systems differ. In D_8 , every reflection squares to the identity. In Q_8 , every noncentral class squares to the nontrivial central element. Hence the weighted reverse fibres of the squaring map differ.

Rational word closure of the squaring and fourth-power pullbacks together with their class-form adjoints gives dimensions 17 and 10. The center and represented-commutant dimensions are

	$\dim A$	$\dim Z(A)$	$\dim A'$
D_8	17	2	2
Q_8	10	2	5.

The dimension identities for a semisimple complex algebra acting on a five-dimensional module have unique solutions

$$(d, m) = (4, 1), (1, 1) \quad \text{and} \quad (d, m) = (3, 1), (1, 2),$$

respectively. This proves the decompositions and the separation. $\square$

Corollary 1.2. *The represented power-map algebra is not a function of the ordinary complex character table, and it is not an isoclinism invariant.*

The same separation persists after adjoining class sizes and element orders. At order 64 the first six pairs of groups with matching character tables, class sizes, and element orders appear; the signature $(\dim A, \dim Z(A), \dim A')$ separates two of them, so the power-map algebra does not factor through these data.

1.2. The Monster. The largest case we compute is the Monster $\mathbb{M}$, with 194 classes and exponent divisible by fifteen primes. Let $\Gamma_{\mathbb{M}}$ be the group of permutations of $\text{Conj}(\mathbb{M})$ that preserve class sizes and commute with every prime-power map; it is the automorphism group of the weighted power system.

Theorem A (Theorem 4.22). $\Gamma_{\mathbb{M}}$ is elementary abelian of rank 14, and

$$\mathcal{P}_{\text{ram}}(\mathbb{M}) = \text{End}_{\mathbb{Q}\Gamma_{\mathbb{M}}}(V_{\mathbb{M}}) \cong M_{170}(\mathbb{Q}) \oplus M_5(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q})^{\oplus 4} \oplus \mathbb{Q}^{\oplus 8},$$

of dimension 28958, with center and commutant of dimension 15. The subgroup of $\Gamma_{\mathbb{M}}$ induced by Galois conjugation of classes has rank 13. A complement is generated by the involution

$$(16B \ 16C)(32A \ 32B),$$

which preserves every class size and every power map but is not an automorphism of the character table.

The equality $\mathcal{P}_{\text{ram}}(\mathbb{M}) = \text{End}_{\mathbb{Q}\Gamma_{\mathbb{M}}}(V_{\mathbb{M}})$ says that the algebra is as large as the symmetry group allows: the weighted power graph of the Monster has no fractional symmetries beyond its automorphisms. The proof is a saturation certificate of 186 orbit vectors (Section 4.3). The complementary involution is a symmetry of the power-map-enriched class table invisible to the character table. It accounts for the one nonprincipal block of $\mathcal{P}_{\text{ram}}(\mathbb{M})$ on which Galois conjugation acts trivially.

1.3. Main results. The general theory behind Theorem A is the following. Write $N = \exp(G)$ and $U_N = (\mathbb{Z}/N)^{\times}$. The results are stated for the group case; Theorem B holds for any finite weighted power system (Definition 2.1).

Theorem B (Structure; Theorems 2.14 and 2.18 and Corollary 2.20). $\mathcal{P}_E(G)$ is a semisimple $\mathbb{Q}$ -algebra equal to its own bicommutant in $\text{End}_{\mathbb{Q}}(V_G)$. Its commutant is the space $\text{Bal}_E(G)$ of balanced kernels on the weighted power graph, so $\mathcal{P}_E(G) = \text{Bal}_E(G)'$, and $\mathcal{P}_E(G) = \text{End}_{\mathbb{Q}}(V_G)$ if and only if the only balanced kernels are the scalar multiples of the identity. For $G \neq 1$ the algebra $\mathcal{P}_{\text{ram}}(G)$ is noncommutative.

Theorem C (Frobenius; Proposition 3.2, Theorems 3.4, 3.9, 3.13 and 3.21, and Corollary 3.5). Let $u \in U_N$. Then Ψ_u is an isometry of V_G commuting with $\mathcal{P}_{\text{ram}}(G)$. As a $\mathbb{Q}[U_N]$ -module,

$$V_G \cong \mathcal{E}(G) = \coprod_{R \in \text{Conj}(G)/U_N} F_R,$$

where F_R is the field generated by the values of the irreducible characters on the classes in the rational class R . For $p \nmid N$ the operator Ψ_p is the Frobenius at p , and

$$\prod_{p \nmid N} \det(I - \Psi_p p^{-s})^{-1} = \zeta_{\mathcal{E}(G)}^{(N)}(s).$$

The factors at $p \mid N$ are recovered from inertia invariants of the good-label operators on each element-order layer, giving the complete Euler product of $\zeta_{\mathcal{E}(G)}$. If the commutant of $\mathcal{P}_{\text{ram}}(G)$ is commutative, every simple factor A_k with center K_k carries a character $\lambda_k : U_N \rightarrow \mu(K_k)$, trivial or quadratic when $K_k = \mathbb{Q}$, and $\mathcal{P}_{\text{ram}}(G) = \mathcal{P}_{\text{Ad}}(G)$.

Theorem D (Census; Theorems 4.2 and 4.8 and Corollary 4.16). For the twenty-six sporadic simple groups, $\mathcal{P}_{\text{ram}}(G)$ is determined explicitly. Every simple factor has Schur index one. The commutant is commutative for twenty-two groups, the exceptions being $O'N$, Fi'_{24} , HN , and $\mathbb{B}$. The center

fields are $\mathbb{Q}$, nine copies of $\mathbb{Q}(\zeta_3)$, and one copy of $\mathbb{Q}(\zeta_5)$; the nonrational centers occur exactly for the six pariah groups $J_1, J_3, J_4, Ru, O'N, Ly$, and each comes from a single rational class of prime size three or five.

1.4. Related constructions. Element-level functional graphs of the power map $x \mapsto x^n$ retain the cycle and rooted-tree structure on all elements, as studied for finite abelian and several nonabelian families by Qureshi and Reis [17]. The present construction retains one vertex per conjugacy class, equips the quotient graph with the nonuniform class measure, and uses several labels simultaneously; the class-form adjoint records every reverse fibre weighted by class size, so two class-power graphs with the same unweighted shape can produce different algebras. The forward operators alone are the linearization of the transformation monoid generated by the τ_n [20, 21]; the algebra studied here is the star closure of that image, and it is contained in the coherent closure of the colored digraph [13, 14, 23, 3, 5] whenever that closure contains the class-measure diagonal and its inverse. GAP's documentation notes that power maps are not in general determined by the matrix of irreducible characters alone [12]; Theorem 1.1 is the algebraic form of that remark.

1.5. Outline. Section 2 defines finite weighted power systems, proves the transfer calculus and Theorem B, and derives the saturation and degeneracy criteria. Section 3 treats the Galois action and proves Theorem C. Section 4 contains the sporadic census, the good-label characters, the rank-one saturation method, and the proof of Theorem A, together with the symmetries invisible to the character table. Section 5 computes the algebra for several infinite families and records a comparison panel of small groups. Section 6 describes the computations, and Section 7 collects open questions. Section A records a dynamical invariant of the prime-power maps that separates the sporadic groups.

2. THE ALGEBRA AND ITS COMMUTANT

2.1. Finite weighted power systems. The finite weighted dynamics that the algebra linearizes is the following.

Definition 2.1. Fix a finite label set E . A finite weighted power system is a tuple

$$\mathcal{X} = (X, \mu, x_0, (\tau_e)_{e \in E}),$$

where X is a finite set, $\mu : X \rightarrow \mathbb{Q}_{>0}$ has total mass one, $x_0 \in X$, and the maps $\tau_e : X \rightarrow X$ commute and fix x_0 . An isomorphism $a : \mathcal{X} \rightarrow \mathcal{Y}$ is a bijection satisfying

$$a(x_0) = y_0, \quad \nu(a(x)) = \mu(x), \quad a\tau_e = \nu_e a \quad (e \in E).$$

Definition 2.2. Call the system multiplicatively labeled if E is a set of positive integers, $1 \in E$, and

$$\tau_m \tau_n = \tau_{mn} \text{ whenever } m, n, mn \in E, \quad \tau_1 = \text{id}. \quad (2.1)$$

This hypothesis is not part of Definition 2.1; it is invoked explicitly where it is used, namely in Proposition 2.11 and throughout Section 3. The structural results of the present section do not require it.

Write $V_X = \text{Map}(X, \mathbb{Q})$, let $\mathcal{B}_X = (\delta_x)_{x \in X}$ be its coordinate basis, and put

$$\langle f, h \rangle_\mu = \sum_{x \in X} \mu(x) f(x) h(x).$$

The pullback $P_e f = f \circ \tau_e$ has coordinate matrix

$$(P_e)_{xy} = \mathbf{1}_{\{y = \tau_e(x)\}}, \quad P_e^* = D_\mu^{-1} P_e^\top D_\mu, \quad D_\mu = \text{diag}(\mu(x)). \quad (2.2)$$

Let $\mathbf{c}_\mu = (c_x)_{x \in X}$ be the primitive positive integral vector proportional to μ , and regard both $\mathbf{c}_\mu = \sum_x c_x \delta_x$ and $\eta_0 = \delta_{x_0}$ as marked vectors.

Definition 2.3. *The fully marked linear object attached to $\mathcal{X}$ is*

$$\mathbf{P}(\mathcal{X}) = (V_X, \mathcal{B}_X, \langle \cdot, \cdot \rangle_\mu, \mathbf{1}, \eta_0, \mathbf{c}_\mu, (P_e)_{e \in E}, \mathcal{P}_E(\mathcal{X})),$$

where

$$\mathcal{P}_E(\mathcal{X}) = \text{Alg}_{\mathbb{Q}}\{I, P_e, P_e^* : e \in E\} \subseteq \text{End}_{\mathbb{Q}}(V_X).$$

The forward subalgebra $\mathcal{S}_E(\mathcal{X}) = \text{Alg}_{\mathbb{Q}}\{I, P_e : e \in E\}$ is commutative, since the τ_e commute; $\mathcal{P}_E(\mathcal{X})$ is its star closure. If $a : \mathcal{X} \rightarrow \mathcal{Y}$ is an isomorphism, then $(U_a f)(y) = f(a^{-1}y)$ carries every part of the marking to the corresponding part, is an isometry, and intertwines P_e^X with P_e^Y and their adjoints. Conversely an isomorphism of fully marked objects permutes their coordinate bases and therefore comes from a unique system isomorphism. The fully marked object thus reconstructs $(X, \mu, x_0, (\tau_e))$ up to unique simultaneous relabeling.

More general factor maps do not act on the star closure. Suppose $q : (X, \mu, \tau_e) \rightarrow (Y, \nu, v_e)$ is a surjection with $q_*\mu = \nu$ and $q\tau_e = v_e q$. Pullback $U_q : V_Y \rightarrow V_X$, $U_q f = f \circ q$, is an isometry and

$$P_e^X U_q = U_q P_e^Y. \quad (2.3)$$

It follows only that $U_q^*(P_e^X)^* = (P_e^Y)^* U_q^*$. The stronger relation $(P_e^X)^* U_q = U_q (P_e^Y)^*$, needed to intertwine the generated adjoint-closed algebras, holds if and only if $\text{im } U_q$ is reducing for P_e^X . Equivalently, for every $x \in X$ and $z \in Y$,

$$\frac{1}{\mu(x)} \sum_{\substack{\tau_e y = x \\ q(y) = z}} \mu(y) = \begin{cases} \frac{\nu(z)}{\nu(q(x))}, & v_e z = q(x), \\ 0, & v_e z \neq q(x). \end{cases} \quad (2.4)$$

General measure-preserving factor maps need not satisfy this reverse-fibre condition.

2.1.1. *The system supplied by a finite group.* Let $\text{Conj}(G) = \{C_1, \dots, C_k\}$, with $C_1 = \{1\}$, and set

$$V_G = \text{Map}(\text{Conj}(G), \mathbb{Q}), \quad \langle f, h \rangle_G = \frac{1}{|G|} \sum_{i=1}^k |C_i| f(C_i) h(C_i). \quad (2.5)$$

For $n \geq 1$, write $\sigma_n(i) = j$ when $g_i^n \in C_j$. Then

$$(P_n)_{ij} = \mathbf{1}_{\{j=\sigma_n(i)\}}, \quad P_n^* = D_G^{-1} P_n^T D_G, \quad D_G = \text{diag} \left(\frac{|C_i|}{|G|} \right). \quad (2.6)$$

Thus G supplies the finite weighted power system

$$\mathcal{C}_E(G) = \left(\text{Conj}(G), \frac{|C|}{|G|}, \{1\}, ([x] \mapsto [x^n])_{n \in E} \right).$$

Its primitive weight vector is the positive class-size vector $(|C_i|)_i$. We define

$$\mathbf{P}(G, E) = \mathbf{P}(\mathcal{C}_E(G)), \quad \mathcal{P}_E(G) = \mathcal{P}_E(\mathcal{C}_E(G)). \quad (2.7)$$

For $N = \exp(G)$, the canonical label set is $E = \pi(N)$, and we omit E from the notation.

A group isomorphism induces an isomorphism of these fully marked objects. More generally, simultaneous relabeling of classes preserving class sizes, the identity class, and all selected power maps is isomorphism at the source-system level.

Proposition 2.4. *If $q : G \twoheadrightarrow Q$ is a quotient, then*

$$\bar{q} : \text{Conj}(G) \longrightarrow \text{Conj}(Q), \quad [g] \longmapsto [q(g)],$$

is a surjective measure-preserving factor map for every power map. Consequently pullback $U_{\bar{q}} : V_Q \rightarrow V_G$ is an isometry and $\Psi_n^G U_{\bar{q}} = U_{\bar{q}} \Psi_n^Q$. It intertwines the full algebras if and only if its image is reducing for every Ψ_n^G .

Proof. The map on classes is well defined, surjective, and commutes with powers. For a class $D \subseteq Q$, its full inverse image has $|\ker q||D|$ elements. Hence the sum of the normalized G -class weights above D is

$$\frac{|\ker q||D|}{|G|} = \frac{|D|}{|Q|}.$$

The remaining assertions are (2.3) and its reverse-fibre criterion. $\square$

A proper subgroup inclusion does not preserve normalized class measure (the identity class has mass $1/|H|$ in H and $1/|G|$ in G), so no proper inclusion is a morphism of weighted power systems.

2.2. Transfer operators. Let $\mathcal{X} = (X, \mu, x_0, (\tau_e)_{e \in E})$ be a finite weighted power system. Write

$$P_e f = f \circ \tau_e, \quad L_e = P_e^*,$$

and let M_a denote multiplication by $a \in V_X$. The reverse-fibre density is

$$r_e(y) = \frac{1}{\mu(y)} \sum_{\tau_e x = y} \mu(x). \quad (2.8)$$

Theorem 2.5. *For $a, b \in V_X$ and $e, f \in E$,*

$$P_e M_a = M_{P_e a} P_e, \quad (2.9)$$

$$L_e((P_e a)b) = a L_e b, \quad (2.10)$$

$$L_e M_b P_e = M_{L_e b}, \quad (2.11)$$

$$L_e P_e = M_{r_e}. \quad (2.12)$$

If $\tau_{ef} = \tau_e \tau_f = \tau_f \tau_e$, then

$$L_e L_f = L_f L_e = L_{ef}, \quad r_{ef} = L_e r_f = L_f r_e. \quad (2.13)$$

Proof. The first identity follows from

$$(P_e(a h))(x) = a(\tau_e x) h(\tau_e x).$$

The adjoint is

$$(L_e b)(y) = \frac{1}{\mu(y)} \sum_{\tau_e x = y} \mu(x) b(x). \quad (2.14)$$

On the fibre $\tau_e x = y$, the function $P_e a$ has value $a(y)$. This gives (2.10) and (2.11). Taking $b = 1$ gives (2.12). The last two identities follow by taking adjoints in $P_e P_f = P_f P_e = P_{ef}$ and applying the result to 1. $\square$

Theorem 2.6. *For a label e , the following conditions are equivalent:*

- (1) P_e is normal;
- (2) P_e is unitary;
- (3) τ_e is a measure-preserving permutation;
- (4) $r_e = 1$ and τ_e is injective.

Consequently

$$\mathcal{P}_E(\mathcal{X}) \text{ is commutative} \iff \text{every } \tau_e \text{ is a measure-preserving permutation.} \quad (2.15)$$

Proof. The matrix $L_e P_e = M_{r_e}$ is diagonal, while

$$(P_e L_e)_{xz} = \begin{cases} \mu(z)/\mu(\tau_e x), & \tau_e z = \tau_e x, \\ 0, & \tau_e z \neq \tau_e x. \end{cases} \quad (2.16)$$

If P_e is normal, (2.16) is diagonal. Every fibre of τ_e then has at most one point, so τ_e is a permutation. For a permutation,

$$r_e(y) = \frac{\mu(\tau_e^{-1}y)}{\mu(y)}.$$

Normality gives $r_e(x) = r_e(\tau_e x)$. This value is constant on each cycle, and its product around the cycle is 1; hence $r_e = 1$. Thus τ_e preserves μ and P_e is unitary. The remaining implications are immediate. Commuting measure-preserving permutations have commuting inverse pullbacks, which proves (2.15). $\square$

Corollary 2.7. *Let G be finite and $N = \exp(G)$. Then*

$$\Psi_n \text{ is normal} \iff \Psi_n \text{ is unitary} \iff (n, N) = 1. \quad (2.17)$$

The good labels generate a commutative semisimple algebra. Every prime $p \mid N$ gives a nonnormal generator; hence $\mathcal{P}_{\text{ram}}(G)$ is noncommutative for $G \neq 1$.

Proof. If $(n, N) = 1$, choose m with $mn \equiv 1 \pmod{N}$. The n -th and m -th power maps are inverse permutations of the conjugacy classes and preserve their sizes. If $(n, N) > 1$, choose $p \mid (n, N)$ and an element x of order p . Then $x^n = 1$, so the power map on conjugacy classes is not injective. Apply Theorem 2.6. $\square$

For a finite group, (2.8) becomes

$$r_n(C) = \frac{\#\{x \in G : x^n \in C\}}{|C|}, \quad r_n(\{1\}) = \#\{x \in G : x^n = 1\}. \quad (2.18)$$

Thus $\Psi_n^* \Psi_n$ is the diagonal operator of class-resolved n -th root counts.

Definition 2.8. *Let $\mathcal{R}_E(\mathcal{X})$ be the smallest unital $\mathbb{Q}$ -subalgebra of V_X stable under every L_e , and put*

$$\mathcal{D}_E(\mathcal{X}) = \{M_a : a \in \mathcal{R}_E(\mathcal{X})\}.$$

Proposition 2.9. *For every finite weighted power system,*

$$\mathcal{D}_E(\mathcal{X}) \subseteq \mathcal{P}_E(\mathcal{X}).$$

Proof. The identity multiplier belongs to $\mathcal{P}_E(\mathcal{X})$. If M_a belongs to it, then

$$M_{L_e a} = L_e M_a P_e \in \mathcal{P}_E(\mathcal{X})$$

by (2.11). The multipliers are closed under sums and products. $\square$

Theorem 2.10. *Let Γ_E be the undirected graph on X with edges $\{x, \tau_e x\}$. If $\mathcal{R}_E(\mathcal{X})$ separates the points of X , then*

$$\mathcal{P}_E(\mathcal{X}) \cong \bigoplus_{C \in \pi_0(\Gamma_E)} \text{End}_{\mathbb{Q}}(\mathbb{Q}^C). \quad (2.19)$$

If Γ_E is connected, then $\mathcal{P}_E(\mathcal{X}) = \text{End}_{\mathbb{Q}}(V_X)$.

Proof. For $x \in X$ and $y \neq x$, choose $a_y \in \mathcal{R}_E(\mathcal{X})$ with $a_y(x) \neq a_y(y)$. Then

$$q_x = \prod_{y \neq x} \frac{a_y - a_y(y)}{a_y(x) - a_y(y)} = \mathbf{1}_{\{x\}}.$$

Hence all coordinate idempotents $E_{xx} = M_{\mathbf{1}_{\{x\}}}$ belong to $\mathcal{P}_E(\mathcal{X})$. If $\tau_e x = y$, then

$$E_{xx} P_e E_{yy} = E_{xy}, \quad E_{yy} L_e E_{xx} = \frac{\mu(x)}{\mu(y)} E_{yx}.$$

Products along paths in Γ_E give every matrix unit within each connected component. Every generator preserves the direct sum over the components. $\square$

2.3. Semisimplicity and the rational bicommutant.

Proposition 2.11. *Let $N = \exp(G)$. Each of the following sets generates the same unital adjoint-closed rational algebra:*

$$\{\Psi_p : p \mid N, p \text{ prime}\}, \quad \{\Psi_{p^a} : p^a \mid N\}, \quad \{\Psi_d : d \mid N\}. \quad (2.20)$$

Here the adjoints of the displayed operators are included in every case. This common algebra is $\mathcal{P}_{\text{ram}}(G)$.

Proof. The identities

$$\Psi_m \Psi_n = \Psi_{mn}, \quad \Psi_1 = I$$

show that $\Psi_{p^a} = \Psi_p^a$ and, for $d = \prod_p p^{a_p}$, that $\Psi_d = \prod_p \Psi_p^{a_p}$. Thus all divisor maps lie in the algebra generated by the prime maps. The reverse inclusion is immediate. Since an adjoint-closed algebra contains the adjoint of every product it contains, adjoining the listed adjoints does not alter this conclusion. $\square$

Example 2.12. *The algebra generated by Ψ_n for every $n \geq 1$ need not equal $\mathcal{P}_{\text{ram}}(G)$. For example, if $G = C_5$, the canonical algebra uses only Ψ_5 , whereas Ψ_2 cyclically permutes the four nonidentity class coordinates. The prime-exponent calculation in Theorem 5.1 shows that the canonical algebra acts scalarly on a three-dimensional complementary summand, while Ψ_2 does not. Hence the all-exponents algebra is strictly larger in this example. Proposition 2.11 concerns the power maps indexed by divisors of $\exp(G)$, not all residue classes modulo the exponent.*

For a subalgebra $A \subseteq \text{End}_F(V)$, write A' for its commutant in $\text{End}_F(V)$.

Lemma 2.13. *Let $F \subseteq K$ be a field extension, let V be a finite-dimensional F -vector space, and let $A \subseteq \text{End}_F(V)$ be a finite-dimensional subalgebra. Under the natural identification $\text{End}_F(V) \otimes_F K \cong \text{End}_K(V \otimes_F K)$, one has*

$$A' \otimes_F K = (A \otimes_F K)'. \quad (2.21)$$

Proof. Choose an F -basis $a_1, \dots, a_r$ of A . Its commutant is the kernel of the F -linear map

$$T : \text{End}_F(V) \longrightarrow \text{End}_F(V)^{\oplus r}, \quad X \longmapsto ([X, a_1], \dots, [X, a_r]).$$

Tensoring the exact sequence $0 \rightarrow \ker T \rightarrow \text{End}_F(V) \xrightarrow{T} \text{End}_F(V)^{\oplus r}$ with the field K preserves exactness. The resulting kernel is the commutant of $A \otimes_F K$. $\square$

Theorem 2.14. *Let X be a finite set, let $\mu : X \rightarrow \mathbb{Q}_{>0}$, and let $(T_e)_{e \in E}$ be any finite family of $\mathbb{Q}$ -linear operators on V_X , with $T_e^* = D_\mu^{-1} T_e^\top D_\mu$ the adjoint for $\langle \cdot, \cdot \rangle_\mu$. Then*

$$A = \text{Alg}_{\mathbb{Q}}\{I, T_e, T_e^* : e \in E\} \subseteq \text{End}_{\mathbb{Q}}(V_X)$$

is a finite-dimensional semisimple $\mathbb{Q}$ -algebra, and in its defining faithful representation on V_X ,

$$A'' = A. \quad (2.22)$$

Over $\mathbb{C}$, there are pairwise inequivalent simple modules $W_1, \dots, W_t$, with $d_j = \dim_{\mathbb{C}} W_j$, and multiplicity spaces M_j , with $m_j = \dim_{\mathbb{C}} M_j$, such that

$$V_X \otimes_{\mathbb{Q}} \mathbb{C} \cong \bigoplus_{j=1}^t W_j \otimes_{\mathbb{C}} M_j, \quad A \otimes_{\mathbb{Q}} \mathbb{C} \cong \bigoplus_{j=1}^t \text{End}_{\mathbb{C}}(W_j) \otimes I_{M_j}. \quad (2.23)$$

In particular,

$$\dim_{\mathbb{C}}(A \otimes \mathbb{C}) = \sum_j d_j^2, \quad \dim_{\mathbb{C}}(A \otimes \mathbb{C})' = \sum_j m_j^2. \quad (2.24)$$

In particular this applies to $A = \mathcal{P}_E(\mathcal{X})$ for any finite weighted power system, and to $A = \mathcal{P}_E(G)$ for any finite group and any exponent set.

Theorem 2.14 uses none of the structure of a finite weighted power system beyond positivity of μ : the T_e need not commute, need not be pullbacks along maps, and no distinguished point is required. The star closure alone gives semisimplicity and bicommutancy. Deterministic commuting maps are what give the balanced-kernel description of the commutant (**Theorem 2.18**), and multiplicative labels are what give the good-label arithmetic of **Section 3**.

Proof. The algebra is finite-dimensional because it is a subalgebra of $\text{End}_{\mathbb{Q}}(V_X)$. Extend scalars and the form $\langle \cdot, \cdot \rangle_{\mu}$ to obtain a Hermitian form on $V_X \otimes_{\mathbb{Q}} \mathbb{C}$, positive definite because $\mu > 0$ pointwise. The complex algebra $A_{\mathbb{C}} = A \otimes_{\mathbb{Q}} \mathbb{C}$ is closed under its Hilbert-space adjoint. Its Jacobson radical J is preserved by the involution because the involution is an anti-automorphism and the radical is invariant under anti-automorphisms. If $x \in J$, then $x^*x \in J$. The ideal J is nilpotent, while x^*x is positive semidefinite, so $x^*x = 0$ and therefore $x = 0$. Hence $A_{\mathbb{C}}$, and consequently A , is semisimple.

The finite-dimensional structure theorem gives the isotypic form (2.23) [7]. On each summand, its commutant is $I_{W_j} \otimes \text{End}_{\mathbb{C}}(M_j)$, whose commutant is $\text{End}_{\mathbb{C}}(W_j) \otimes I_{M_j}$. Thus $A''_{\mathbb{C}} = A_{\mathbb{C}}$. Applying **Lemma 2.13** twice gives

$$A'' \otimes_{\mathbb{Q}} \mathbb{C} = A''_{\mathbb{C}} = A_{\mathbb{C}} = A \otimes_{\mathbb{Q}} \mathbb{C}.$$

Since $A \subseteq A''$ and scalar extension preserves dimensions, this forces $A = A''$. The dimension identities follow from the two displayed block actions. The final sentence is the case $T_e = P_e$ of (2.2). $\square$

Proposition 2.15. *For every field extension $K/\mathbb{Q}$,*

$$\mathcal{P}_E(G) \otimes_{\mathbb{Q}} K = \text{Alg}_K\{I, \Psi_n, \Psi_n^* : n \in E\} \subseteq \text{End}_K(V_G \otimes_{\mathbb{Q}} K),$$

and

$$\mathcal{P}_E(G)' \otimes_{\mathbb{Q}} K = (\mathcal{P}_E(G) \otimes_{\mathbb{Q}} K)'.$$

Thus extension of scalars preserves the represented algebra and its commutant. The center fields, Galois orbits, and Schur indices give the descent data that recover the rational form from its complex blocks.

Proof. The first identity follows because the same rational generators span both sides after scalar extension. The second is **Lemma 2.13**. The final statement is the usual distinction between a semisimple rational algebra and its split complex form. $\square$

Direct products exhibit a natural tensor relation, but the power maps on the two factors are synchronized by the same exponent.

Proposition 2.16. *For finite groups G, H and any exponent set E , the canonical identification*

$$V_{G \times H} \cong V_G \otimes_{\mathbb{Q}} V_H$$

satisfies

$$\Psi_n^{G \times H} = \Psi_n^G \otimes \Psi_n^H, \quad (\Psi_n^{G \times H})^* = (\Psi_n^G)^* \otimes (\Psi_n^H)^*. \quad (2.25)$$

Consequently,

$$\mathcal{P}_E(G \times H) \subseteq \mathcal{P}_E(G) \otimes_{\mathbb{Q}} \mathcal{P}_E(H) \quad (2.26)$$

as represented algebras. For canonical exponent sets, take $E = \pi(\exp(G \times H))$. If G and H have the same prime divisors in their exponents, the right-hand side is $\mathcal{P}_{\text{ram}}(G) \otimes \mathcal{P}_{\text{ram}}(H)$.

Proof. The conjugacy classes of $G \times H$ are the products $C_i \times D_j$, their sizes multiply, and $(g, h)^n = (g^n, h^n)$. Hence the power matrix and class-measure matrix are

$$P_n^{G \times H} = P_n^G \otimes P_n^H, \quad D_{G \times H} = D_G \otimes D_H.$$

Formula (2.2) gives the second identity in (2.25). Every generator on the left of (2.26) therefore belongs to the tensor product algebra on the right. $\square$

Containment in (2.26) can be strict. The left-hand algebra is generated by the synchronized tensors $\Psi_n^G \otimes \Psi_n^H$ and their adjoints. Tensor-product equality holds if and only if these synchronized generators recover the separate operators $\Psi_n^G \otimes I$ and $I \otimes \Psi_n^H$; strict examples occur when they do not.

2.4. Balanced kernels. The commutant has an intrinsic description directly on the weighted colored digraph. Write an operator $S \in \text{End}_{\mathbb{Q}}(V_X)$ in coordinate form

$$(Sf)(x) = \sum_{y \in X} s(x, y)f(y).$$

Definition 2.17. Let $\text{Bal}(\mathcal{X})$ be the rational vector space of kernels $s : X \times X \rightarrow \mathbb{Q}$ satisfying, for every $e \in E$ and $x, z \in X$,

$$s(\tau_e x, z) = \sum_{\tau_e y = z} s(x, y), \quad (2.27)$$

$$\frac{\mu(z)}{\mu(\tau_e z)} s(x, \tau_e z) = \frac{1}{\mu(x)} \sum_{\tau_e y = x} \mu(y) s(y, z). \quad (2.28)$$

Composition is ordinary matrix multiplication.

These kernels balance the colored forward fibres and their weighted reverse fibres.

Theorem 2.18. For every finite weighted power system,

$$\mathcal{P}_E(\mathcal{X})' = \text{Bal}(\mathcal{X}). \quad (2.29)$$

Consequently

$$\mathcal{P}_E(\mathcal{X}) = \text{Bal}(\mathcal{X})' \quad (2.30)$$

in its defining representation. In particular the algebra is determined by the fractional symmetries of the weighted colored power graph.

Proof. Matrix multiplication gives

$$(SP_e)_{xz} = \sum_{\tau_e y = z} s(x, y), \quad (P_e S)_{xz} = s(\tau_e x, z),$$

so $SP_e = P_e S$ is (2.27). From $(P_e^*)_{yz} = \mu(z)\mu(y)^{-1}\mathbf{1}_{\{y=\tau_e z\}}$, one obtains

$$(SP_e^*)_{xz} = \frac{\mu(z)}{\mu(\tau_e z)} s(x, \tau_e z), \quad (P_e^* S)_{xz} = \frac{1}{\mu(x)} \sum_{\tau_e y = x} \mu(y) s(y, z).$$

Thus $SP_e^* = P_e^* S$ is (2.28). Commuting with all generators is equivalent to commuting with their generated algebra, proving (2.29). Equation (2.30) is the rational double-commutant theorem. $\square$

Every source automorphism gives a permutation kernel in $\text{Bal}(\mathcal{X})$. Automorphisms alone usually do not describe the commutant: arbitrary rational balanced kernels are allowed. For a group of prime exponent with k classes, the decomposition in Theorem 5.1 gives the explicit formula

$$\mathcal{P}_{\text{ram}}(G)' \cong \mathbb{Q} \oplus M_{k-2}(\mathbb{Q}), \quad \dim_{\mathbb{Q}} \mathcal{P}_{\text{ram}}(G)' = 1 + (k-2)^2.$$

This is a family-level commutant calculation rather than a matrix nullspace computation.

Between the fully marked object and the abstract algebra there are two further levels of forgetting: the operator-marked object $(V, \langle \cdot, \cdot \rangle, \mathbf{1}, \eta_0, (P_e))$, which retains the labeled generators but not the coordinate basis, and the unmarked represented algebra $(A \subseteq \text{End}_{\mathbb{Q}}(V), V, \langle \cdot, \cdot \rangle)$. Each arrow in this tower is strict over $\mathbb{Q}$: the last by the prime-exponent family, where C_3 and C_5 share the abstract algebra $M_2(\mathbb{Q}) \oplus \mathbb{Q}$ but have scalar factors of different multiplicity; the others by explicit small systems, which we omit.

2.5. Saturation and two degeneracy criteria. The balanced-kernel theorem converts three questions about the algebra into questions about balanced kernels on the power graph: when enlarging the label set changes nothing, when the algebra degenerates to everything, and when the commutant is commutative. All three are consequences of Theorem 2.18 together with the rational double commutant.

Theorem 2.19. *Let $E \subseteq F$ be finite label sets for $\mathcal{X}$. Then $\mathcal{P}_E(\mathcal{X}) \subseteq \mathcal{P}_F(\mathcal{X})$, and*

$$\mathcal{P}_E(\mathcal{X}) = \mathcal{P}_F(\mathcal{X}) \iff \text{every } P_f, f \in F \setminus E, \text{ commutes with every } s \in \text{Bal}_E(\mathcal{X}), \quad (2.31)$$

where Bal_E is the balanced-kernel algebra of Definition 2.17 formed with the label set E . We call E saturated in F when this holds.

Proof. The containment is clear. For the criterion, $\mathcal{P}_F$ is generated over $\mathcal{P}_E$ by the operators P_f and P_f^* with $f \in F \setminus E$, so equality holds if and only if every such P_f lies in $\mathcal{P}_E$; the adjoints then follow because $\mathcal{P}_E$ is adjoint closed. By Theorem 2.14 and Theorem 2.18,

$$\mathcal{P}_E = (\mathcal{P}_E)'' = \text{Bal}_E(\mathcal{X})',$$

and membership in the commutant of Bal_E is the stated commutation condition. $\square$

Corollary 2.20. *$\mathcal{P}_E(\mathcal{X}) = \text{End}_{\mathbb{Q}}(V_X)$ if and only if $\text{Bal}_E(\mathcal{X}) = \mathbb{Q}$, that is, if and only if the only balanced kernels are the scalar multiples of the identity kernel.*

Proof. By Theorem 2.18 and the double commutant, $\mathcal{P}_E = \text{Bal}'_E$, and a subalgebra of $\text{End}_{\mathbb{Q}}(V_X)$ equals $\text{End}_{\mathbb{Q}}(V_X)$ if and only if its commutant is the scalars. $\square$

Corollary 2.21. *The defining module V_X is multiplicity free over $\mathcal{P}_E(\mathcal{X}) \otimes \mathbb{C}$ if and only if $\text{Bal}_E(\mathcal{X})$ is commutative, and in that case $\dim_{\mathbb{Q}} Z(\mathcal{P}_E) = \dim_{\mathbb{Q}} \text{Bal}_E(\mathcal{X})$.*

Proof. With notation as in (2.23), the commutant after scalar extension is $\bigoplus_j \text{End}_{\mathbb{C}}(M_j)$, which is commutative if and only if every $m_j = 1$. Commutativity is preserved and reflected by scalar extension, so it may be tested over $\mathbb{Q}$. When all $m_j = 1$ the commutant and the center both have dimension t . $\square$

Throughout, “ V_G is multiplicity free” means the condition of Corollary 2.21: the commutant $\mathcal{P}_{\text{ram}}(G)'$ is commutative.

First, Theorem 2.19 explains Example 2.12: for $G = C_5$ the label set $E = \{5\}$ is not saturated in $\{2, 5\}$, because $\text{Bal}_{\{5\}}$ contains the full matrix algebra of the three-dimensional complementary summand, and the four-cycle Ψ_2 does not commute with all of it. Hence $\mathcal{P}_{\text{ram}}(C_5) \subsetneq \mathcal{P}_{\text{Ad}}(C_5)$, with dimensions 5 and 7. Second, by Corollary 2.20 the equality $\dim A = k(G)^2$ holds when the weighted colored power graph admits no balanced kernel beyond the scalars. The rows of Theorem 5.9 with $\dim A = k(G)^2$ therefore record rigidity of that graph.

3. GOOD LABELS, THE UNIT GROUP, AND RATIONAL CHARACTERS

Throughout, G is a finite group, $N = \exp(G)$, and

$$U_N = (\mathbb{Z}/N)^{\times}.$$

We call $n \geq 1$ a *good label* when $\gcd(n, N) = 1$ and a *ramified label* when every prime divisor of n divides N .

3.1. The unit group acts by isometries.

Lemma 3.1. *Let n be a good label and let m satisfy $mn \equiv 1 \pmod{N}$. Then:*

- (1) Ψ_n depends only on the residue $n \pmod{N}$;
- (2) $\sigma_n : \text{Conj}(G) \rightarrow \text{Conj}(G)$ is a bijection with inverse σ_m , and $|\sigma_n(C)| = |C|$ for every class C ;
- (3) Ψ_n is orthogonal for the class form (2.5), and

$$\Psi_n^* = \Psi_n^{-1} = \Psi_m; \quad (3.1)$$

- (4) the assignment $u \mapsto \Psi_u$ is a group homomorphism

$$\Psi : U_N \longrightarrow O(V_G, \langle \cdot, \cdot \rangle_G).$$

Proof. (1) For $x \in G$ the order $|x|$ divides N , so x^n depends only on $n \pmod{|x|}$, hence only on $n \pmod{N}$.

(2) Since $x^N = 1$ and $nm \equiv 1 \pmod{N}$, we get $(x^n)^m = x^{nm} = x$ for every $x \in G$. Hence $\sigma_m \sigma_n = \sigma_n \sigma_m = \text{id}$ and σ_n is a bijection. For the class sizes, fix a class C and consider $\theta : C \rightarrow \sigma_n(C)$, $\theta(x) = x^n$. It is well defined because C generates a single class of n th powers. It is injective: if $x^n = y^n$ with $x, y \in C$ then $x = (x^n)^m = (y^n)^m = y$. It is surjective: if $z \in \sigma_n(C)$ then $z^m \in \sigma_m \sigma_n(C) = C$ and $\theta(z^m) = z^{mn} = z$. Hence $|C| = |\sigma_n(C)|$.

- (3) By (2.6), Ψ_n is the permutation operator of σ_n . Using (2),

$$\langle \Psi_n f, \Psi_n h \rangle_G = \frac{1}{|G|} \sum_C |C| f(\sigma_n C) h(\sigma_n C) = \frac{1}{|G|} \sum_{C'} |\sigma_n^{-1} C'| f(C') h(C') = \langle f, h \rangle_G,$$

because $|\sigma_n^{-1} C'| = |C'|$. Thus Ψ_n is orthogonal and $\Psi_n^* = \Psi_n^{-1}$. Finally $\Psi_m \Psi_n = \Psi_{mn} = \Psi_1 = I$ by (1) and multiplicativity, which gives (3.1).

- (4) Immediate from (1) and (3). □

Proposition 3.2. *For every $u \in U_N$ we have $\Psi_u \in \mathcal{P}_{\text{ram}}(G)'$. Consequently $\Psi(U_N)$ is a finite abelian group of isometries inside $\mathcal{P}_{\text{ram}}(G)'$.*

Proof. Let $d \mid N$. Multiplicativity gives $\Psi_u \Psi_d = \Psi_{ud} = \Psi_{du} = \Psi_d \Psi_u$. Taking adjoints of this identity and using (3.1),

$$\Psi_d^* \Psi_{u^{-1}} = \Psi_{u^{-1}} \Psi_d^*.$$

As $u \mapsto u^{-1}$ is a bijection of U_N , the operator Ψ_u commutes with every Ψ_d^* as well. By Proposition 2.11 the operators Ψ_d, Ψ_d^* with $d \mid N$ generate $\mathcal{P}_{\text{ram}}(G)$, and an operator commuting with a generating set commutes with the generated algebra. □

Proposition 3.3. *Every $n \geq 1$ factors uniquely as $n = n_r n_g$ with n_r a ramified label and n_g a good label, and then*

$$\Psi_n = \Psi_{n_r} \Psi_{n_g}, \quad \Psi_{n_r} \in \mathcal{P}_{\text{ram}}(G), \quad \Psi_{n_g} = \Psi_u, \quad u = n_g \pmod{N} \in U_N. \quad (3.2)$$

Hence

$$\mathcal{P}_{\text{Ad}}(G) = \text{Alg}_{\mathbb{Q}}(\mathcal{P}_{\text{ram}}(G) \cup \Psi(U_N)). \quad (3.3)$$

Proof. Let n_r be the product of the prime powers of n at primes dividing N and $n_g = n/n_r$; uniqueness is clear. Multiplicativity gives the first identity, and Proposition 2.11 places Ψ_{n_r} in $\mathcal{P}_{\text{ram}}(G)$. Since $\Psi(U_N)$ is closed under adjoints by (3.1), and $\mathcal{P}_{\text{Ad}}(G)$ is generated by all Ψ_n and their adjoints, (3.3) follows. □

3.2. Rational classes and fields of character values. Identify U_N with $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ by $u \mapsto (\zeta_N \mapsto \zeta_N^u)$. For a rational class $R \in \text{Conj}(G)/U_N$ put

$$\Gamma_R = \{u \in U_N : \sigma_u(C) = C \text{ for some, equivalently every, } C \in R\}, \quad F_R = \mathbb{Q}(\zeta_N)^{\Gamma_R},$$

the stabilizer being independent of the choice of C because U_N is abelian. The field F_R is a classical object: it is the field of character values on R . Indeed, for $g \in C \in R$ and $\chi \in \text{Irr}(G)$, the Galois element u sends $\chi(g)$ to $\chi(g^u)$, so u fixes every $\chi(g)$ if and only if $\chi(g^u) = \chi(g)$ for all χ , that is, if and only if g^u is conjugate to g [15]. Hence $F_R = \mathbb{Q}(\chi(g) : \chi \in \text{Irr}(G))$, and the étale algebra below is the product of the fields of values of the rational classes. Set

$$\mathcal{E}(G) = \prod_{R \in \text{Conj}(G)/U_N} F_R. \quad (3.4)$$

Theorem 3.4. *There is an isomorphism of $\mathbb{Q}[U_N]$ -modules*

$$V_G \cong \bigoplus_R \mathbb{Q}[U_N/\Gamma_R] \cong \bigoplus_R F_R = \mathcal{E}(G), \quad (3.5)$$

under which Ψ_u acts as the field automorphism determined by u on each factor. In particular $[F_R : \mathbb{Q}] = |R|$ and $\dim_{\mathbb{Q}} \mathcal{E}(G) = k(G)$.

Proof. By Lemma 3.1, Ψ_u is the permutation operator of σ_u in the basis $(\delta_C)_C$, so V_G is the permutation module $\mathbb{Q}[\text{Conj}(G)]$ for U_N . Decomposing $\text{Conj}(G)$ into orbits gives $V_G \cong \bigoplus_R \mathbb{Q}[R]$, and $\mathbb{Q}[R] \cong \mathbb{Q}[U_N/\Gamma_R]$ because a transitive action with point stabilizer Γ_R is isomorphic to the coset action. Thus $|R| = [U_N : \Gamma_R] = [F_R : \mathbb{Q}]$.

By the normal basis theorem $\mathbb{Q}(\zeta_N) \cong \mathbb{Q}[U_N]$ as $\mathbb{Q}[U_N]$ -modules. Taking Γ_R -invariants on both sides and using that U_N is abelian, so that Γ_R -invariants of $\mathbb{Q}[U_N]$ are spanned by the Γ_R -coset sums, gives $F_R = \mathbb{Q}(\zeta_N)^{\Gamma_R} \cong \mathbb{Q}[U_N/\Gamma_R]$ as $\mathbb{Q}[U_N]$ -modules. Summing over R and adding the class counts gives $\dim_{\mathbb{Q}} \mathcal{E}(G) = \sum_R |R| = k(G)$. $\square$

Corollary 3.5. *For $\text{Re } s > 1$,*

$$\prod_{p \nmid N} \det(I - \Psi_p p^{-s})^{-1} = \prod_R \zeta_{F_R}^{(N)}(s) = \zeta_{\mathcal{E}(G)}^{(N)}(s), \quad (3.6)$$

the superscript denoting omission of the Euler factors at the primes dividing N .

Proof. Under (3.5) the representation of $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ on V_G is $\bigoplus_R \text{Ind}_{\Gamma_R}^{U_N} \mathbf{1}$, whose Artin L -function is $\prod_R \zeta_{F_R}$ by Artin induction [22, §VII]. For $p \nmid N$ the class of p in U_N is the Frobenius at p , and $\Psi_p = \Psi_{p \bmod N}$ by Lemma 3.1(1), so $\det(I - \Psi_p p^{-s})^{-1}$ is the Euler factor at p of that L -function. $\square$

Theorem 3.6. *Let $\widehat{U_N} = \text{Hom}(U_N, \mathbb{C}^\times)$, and for $\chi \in \widehat{U_N}$ put*

$$m_G(\chi) = \#\{R \in \text{Conj}(G)/U_N : \Gamma_R \subseteq \ker \chi\}. \quad (3.7)$$

Then the good-label representation has the exact complex decomposition

$$V_G \otimes_{\mathbb{Q}} \mathbb{C} \cong \bigoplus_{\chi \in \widehat{U_N}} \chi^{\oplus m_G(\chi)}, \quad (3.8)$$

and its Euler product factors as

$$\prod_{p \nmid N} \det(I - \Psi_p p^{-s})^{-1} = \prod_{\chi \in \widehat{U_N}} L^{(N)}(s, \chi)^{m_G(\chi)}. \quad (3.9)$$

Here χ is viewed as the corresponding Dirichlet character of conductor dividing N , and $L^{(N)}$ denotes the Euler product with the primes dividing N omitted. Thus the rational-class stabilizers determine every Dirichlet factor and its multiplicity without a Wedderburn computation.

Proof. By Theorem 3.4, the summand attached to R is $\text{Ind}_{\Gamma_R}^{U_N} \mathbf{1}$. Since U_N is abelian, Frobenius reciprocity gives

$$\dim_{\mathbb{C}} \text{Hom}_{U_N}(\chi, \text{Ind}_{\Gamma_R}^{U_N} \mathbf{1}) = \dim_{\mathbb{C}} \text{Hom}_{\Gamma_R}(\chi|_{\Gamma_R}, \mathbf{1}) = \begin{cases} 1, & \Gamma_R \subseteq \ker \chi, \\ 0, & \text{otherwise.} \end{cases}$$

Summing over R proves (3.8) and (3.7). For $p \nmid N$, the operator Ψ_p acts on the χ -line by $\chi(p)$; taking determinants and multiplying the factors $(1 - \chi(p)p^{-s})^{-1}$ over the good primes proves (3.9). $\square$

Corollary 3.7. *The field F_R is totally real if and only if $-1 \in \Gamma_R$, that is, if and only if the classes in R are real. Writing $(r_1(R), r_2(R))$ for the signature of F_R , the completed product $\prod_R \zeta_{F_R}$ has archimedean factor*

$$\Gamma_{\mathbf{R}}(s) \sum_R (r_1(R) + r_2(R)) \Gamma_{\mathbf{R}}(s+1) \sum_R r_2(R), \quad \Gamma_{\mathbf{R}}(s) = \pi^{-s/2} \Gamma(s/2), \quad (3.10)$$

and conductor $\prod_R |d_{F_R}|$.

Proof. Complex conjugation on $\mathbb{Q}(\zeta_N)$ is Ψ_{-1} , and F_R is fixed pointwise by it if and only if $-1 \in \Gamma_R$; a subfield of a cyclotomic field is either totally real or totally imaginary, so this is the criterion for total reality. By Lemma 3.1(2) the condition $\sigma_{-1}(C) = C$ says that $C = C^{-1}$. The displayed factor is $\prod_R \Gamma_{\mathbf{R}}(s)^{r_1(R)} \Gamma_{\mathbf{C}}(s)^{r_2(R)}$ with $\Gamma_{\mathbf{C}}(s) = \Gamma_{\mathbf{R}}(s) \Gamma_{\mathbf{R}}(s+1)$. $\square$

For the Monster, $k(\mathbb{M}) = 194$ with 172 rational classes (Proposition 4.17), so 150 are singletons and 22 have two elements, the latter imaginary quadratic by (4.4). Then $\sum_R (r_1 + r_2) = 172$, $\sum_R r_2 = 22$, and (3.10) reads

$$\Gamma_{\mathbf{R}}(s)^{172} \Gamma_{\mathbf{R}}(s+1)^{22}.$$

Here every F_R has degree at most two. For J_1 the orbit of size three at element order 19 has Γ_R of order six in $U_{19} \cong \mathbb{Z}/18$, and that subgroup contains the involution, so F_R is a totally real cubic field contributing $\Gamma_{\mathbf{R}}(s)^3$. Its block center $\mathbb{Q}(\zeta_3)$ in Theorem 4.14 is another field: $\mathbb{Q}[U_N/\Gamma_R] \cong \mathbb{Q} \oplus W_R$ with F_R on the left and $\mathbb{Q}(\zeta_3) = \text{End}_{\mathbb{Q}[U_N]}(W_R)$ on the right.

3.3. Ramified labels.

Theorem 3.8. *Let $p \mid N$, let $b = v_p(N)$, and let $\text{Conj}(G)_{p'}$ be the set of classes of elements of order prime to p . Then σ_p restricts to a bijection of $\text{Conj}(G)_{p'}$, the span*

$$W_p = \text{span}_{\mathbb{Q}}(\delta_C : C \notin \text{Conj}(G)_{p'})$$

is Ψ_p -invariant with $\Psi_p^b|_{W_p} = 0$, and Ψ_p induces on V_G/W_p the permutation operator of $\sigma_p|_{\text{Conj}(G)_{p'}}$. Consequently the characteristic polynomial of Ψ_p is

$$X^{\dim W_p} \prod_{\gamma} (X^{\ell(\gamma)} - 1),$$

the product being over the cycles γ of $\sigma_p|_{\text{Conj}(G)_{p'}}$, and every nonzero eigenvalue of Ψ_p is a root of unity.

Proof. If $|x| = m$ with $p \nmid m$ then $|x^p| = m$, so σ_p maps $\text{Conj}(G)_{p'}$ into itself; it is injective there, since choosing m' with $pm' \equiv 1 \pmod{m}$ recovers x from x^p inside $\langle x \rangle$, hence bijective. Conversely if $p \mid |x|$ then $p \mid |x^p|$ unless $|x| = p$, in which case $x^p = 1$; so for $C \notin \text{Conj}(G)_{p'}$ every class in $\sigma_p^{-1}(C)$ also lies outside $\text{Conj}(G)_{p'}$. By (2.6), $\Psi_p \delta_C = \sum_{\sigma_p(C')=C} \delta_{C'}$, so W_p is Ψ_p -invariant. Moreover $|x^{p^b}|$ is prime to p for every $x \in G$, so $\sigma_p^b(\text{Conj}(G)) \subseteq \text{Conj}(G)_{p'}$ and $\sigma_p^{-b}(C) = \emptyset$ for $C \notin \text{Conj}(G)_{p'}$; hence $\Psi_p^b|_{W_p} = 0$.

The images of the δ_C with $C \in \text{Conj}(G)_{p'}$ form a basis of V_G/W_p , and Ψ_p sends δ_C to $\delta_{\sigma_p^{-1}(C) \cap \text{Conj}(G)_{p'}}$ modulo W_p , which is the permutation operator of the bijection of the first paragraph. The characteristic polynomial is the product of those of $\Psi_p|_{W_p}$ and of the induced map, and a permutation of cycle type $(\ell(\gamma))_{\gamma}$ has characteristic polynomial $\prod_{\gamma} (X^{\ell(\gamma)} - 1)$. $\square$

3.4. Local factors at ramified primes. The good primes give the Euler factors of $\zeta_{\mathcal{E}(G)}$ away from N . The factors at $p \mid N$ are determined, as for any abelian étale algebra, by Frobenius acting on inertia invariants; here both are built from good-label operators restricted to element-order layers.

For $m \mid N$, let X_m be the classes of elements of order m and put $V_m = \text{Map}(X_m, \mathbb{Q})$. The action of U_N on X_m factors through $U_m = (\mathbb{Z}/m)^\times$. We write $\Psi_{u,m}$ for the resulting operator; a lift of u to U_N gives the same operator on V_m .

Fix a prime p . If $p \nmid m$, set $I_{p,m} = 1$ and $\kappa_{p,m} = p \bmod m$. If $m = p^a m'$, with $a > 0$ and $p \nmid m'$, put

$$I_{p,m} = \ker(U_m \longrightarrow U_{m'})$$

and choose $\kappa_{p,m} \in U_m$ such that

$$\kappa_{p,m} \equiv p \pmod{m'}, \quad \kappa_{p,m} \equiv 1 \pmod{p^a}. \quad (3.11)$$

Let

$$\Pi_{p,m} = \frac{1}{|I_{p,m}|} \sum_{u \in I_{p,m}} \Psi_{u,m}, \quad \Phi_{p,m} = \Pi_{p,m} \Psi_{\kappa_{p,m},m} \Pi_{p,m} \big|_{V_m^{I_{p,m}}}. \quad (3.12)$$

The first operator is the orthogonal projection onto $V_m^{I_{p,m}}$. The second is independent of the choice in (3.11), since two choices differ by an element of $I_{p,m}$.

Define

$$\mathcal{L}_p(G, T) = \prod_{m \mid N} \det(I - \Phi_{p,m} T \big| V_m^{I_{p,m}}). \quad (3.13)$$

Theorem 3.9. *For $\text{Re } s > 1$,*

$$\zeta_{\mathcal{E}(G)}(s) = \prod_R \zeta_{F_R}(s) = \prod_p \mathcal{L}_p(G, p^{-s})^{-1}. \quad (3.14)$$

If $p \nmid N$, then

$$\mathcal{L}_p(G, T) = \det(I - \Psi_p T \big| V_G).$$

If $p \mid N$, the factors with $p \nmid m$ are the Frobenius determinants on the p -regular order layers, and

$$C_p(G, T) = \prod_{\substack{m \mid N \\ p \nmid m}} \det(I - \Phi_{p,m} T \big| V_m^{I_{p,m}}) \quad (3.15)$$

is the contribution of the p -singular layers.

Proof. Every U_N -orbit R lies in one X_m . On $\mathbb{Q}[R]$, the local Artin factor of F_R at p is

$$\det(I - \text{Frob}_p T \big| \mathbb{Q}[R]^{I_{p,m}})^{-1}.$$

For $p \nmid m$, inertia is trivial and $p \bmod m$ acts as Frobenius. For $p \mid m$, (3.11) represents Frobenius modulo inertia, so $\Phi_{p,m}$ is its action on the inertia invariants. Multiplication over the orbits in X_m , then over $m \mid N$, gives $\mathcal{L}_p(G, T)^{-1}$. Multiplication over p gives (3.14). If $p \nmid N$, all inertia groups are trivial and the direct sum of the $\Psi_{p,m}$ is Ψ_p . $\square$

Corollary 3.10. *The class sizes together with all power maps, that is, the ramified eidetic object $\mathbf{E}_{\text{ram}}(G)$, determine every local factor of $\zeta_{\mathcal{E}(G)}$.*

Corollary 3.11. *The degree of $C_p(G, T)$ is*

$$\delta_p(G) = \sum_{\substack{m \mid N \\ p \nmid m}} \#(I_{p,m} \setminus X_m).$$

If $r_p(G)$ is the number of p -singular rational classes and $c_p(G)$ the number of p -singular conjugacy classes, then

$$r_p(G) \leq \delta_p(G) \leq c_p(G).$$

Proof. The fixed space of a permutation group on X_m has dimension equal to its number of orbits. Since $I_{p,m} \leq U_m$, its orbit partition refines the rational-class partition and coarsens the partition into points. $\square$

For the Monster, the ramified degrees are

p	2	3	5	7	11	13	17	19	23	29	31	41	47	59	71
δ_p	130	90	44	28	11	12	5	4	5	2	3	1	2	1	1

At $p = 2, 3$,

$$122 < 130 < 133, \quad 88 < 90 < 93.$$

3.5. Good-label representations.

Theorem 3.12. *Let $A = \mathcal{P}_{\text{ram}}(G)$ and $N = \exp(G)$. Choose representatives $S_1, \dots, S_t$ for the simple rational A -modules occurring in V_G , and write*

$$D_j = \text{End}_A(S_j), \quad M_j = \text{Hom}_A(S_j, V_G).$$

Then evaluation gives an A -module decomposition

$$V_G \cong \bigoplus_{j=1}^t S_j \otimes_{D_j} M_j, \quad (3.16)$$

and restriction to the isotypic components gives

$$A' \cong \bigoplus_{j=1}^t \text{End}_{D_j}(M_j). \quad (3.17)$$

For every j , the good labels induce a canonical representation

$$\rho_j : U_N \longrightarrow \text{End}_{D_j}(M_j)^\times, \quad \rho_j(u)(f) = \Psi_u \circ f, \quad (3.18)$$

with the following properties.

- (1) *The image is finite and $\rho_j(u)^\dagger = \rho_j(u^{-1})$, where $\dagger$ is the positive involution on $\text{End}_{D_j}(M_j)$ induced by the class form.*
- (2) *Every complex eigenvalue of every $\rho_j(u)$ is a root of unity, and every rational characteristic polynomial of $\rho_j(u)$ is a product of cyclotomic polynomials.*
- (3) *The action on the j -th component is scalar if and only if $\rho_j(U_N) \subseteq Z(\text{End}_{D_j}(M_j))^\times$. In that event it defines a character $\lambda_j : U_N \rightarrow \mu(K_j)$, where $K_j = Z(D_j)$.*
- (4) *If $D_j = K_j$ and $\dim_{K_j} M_j = 1$, the action is automatically scalar. Thus the split multiplicity-one theorem below is the rank-one specialization of (3.18).*
- (5) *Put $a_j = \dim_{D_j} S_j$. For every $u \in U_N$,*

$$\#\{C \in \text{Conj}(G) : \sigma_u(C) = C\} = \sum_{j=1}^t a_j \text{Tr}_{\mathbb{Q}}(\rho_j(u) \mid M_j). \quad (3.19)$$

Proof. The algebra A is semisimple by Theorem 2.14. The rational double-centralizer theorem therefore gives (3.16) and (3.17). Explicitly, S_j is a right D_j -module through evaluation by A -endomorphisms, M_j is the corresponding left D_j -multiplicity module, and the evaluation map

$$S_j \otimes_{D_j} \text{Hom}_A(S_j, V_G) \longrightarrow V_G, \quad s \otimes f \longmapsto f(s),$$

identifies the source with the S_j -isotypic component.

Every Ψ_u commutes with A by Proposition 3.2, so postcomposition preserves $\text{Hom}_A(S_j, V_G)$ and is D_j -linear. The equality $\Psi_{uv} = \Psi_u \Psi_v$ proves that ρ_j is a representation. Because U_N is finite, its image is finite. The identity $\Psi_u^* = \Psi_{u^{-1}}$ of (3.1) restricts under the double-centralizer identification to $\rho_j(u)^\dagger = \rho_j(u^{-1})$, proving (1).

If u has order e in U_N , then $\rho_j(u)^e = 1$. In characteristic zero the minimal polynomial divides $x^e - 1$, which is square-free, so $\rho_j(u)$ is semisimple with roots of unity as eigenvalues. Its rational characteristic polynomial is therefore a product of cyclotomic polynomials. This proves (2).

For (3),

$$Z(\text{End}_{D_j}(M_j)) = Z(D_j) = K_j,$$

and every finite subgroup of $K_j^\times$ lies in $\mu(K_j)$. If $D_j = K_j$ and $\dim_{K_j} M_j = 1$, then $\text{End}_{K_j}(M_j) = K_j$, proving (4).

Finally, Ψ_u is the permutation operator of the u -power action on conjugacy classes, so its rational trace is the number of fixed classes. Choose a right D_j -basis of S_j of size a_j . On $S_j \otimes_{D_j} M_j$, the operator induced by Ψ_u is the direct sum of a_j copies of $\rho_j(u)$ as a rational linear operator. Taking traces in (3.16) gives (3.19). $\square$

For $m_j > 1$, the good labels define a finite $\dagger$ -unitary representation in $\text{End}_{D_j}(M_j)$. A scalar character exists if its image lies in the center.

3.6. Rational characters on the isotypic components. Write $A = \mathcal{P}_{\text{ram}}(G)$. Proposition 3.2 says that the good labels live in A' ; the structure theory of Section 2.3 therefore constrains them as soon as we know A' . The strongest statement occurs in the multiplicity-free case, which by Corollary 2.21 is the case A' commutative.

Theorem 3.13. *Suppose V_G is multiplicity free over $A = \mathcal{P}_{\text{ram}}(G)$. Let $E_1, \dots, E_t$ be the primitive idempotents of $Z(A)$, let $AE_k \cong M_{r_k}(D_k)$ be the corresponding simple factors, and let $K_k = Z(AE_k)$ be their center fields. Then $A' = Z(A)$, and for every $u \in U_N$*

$$\Psi_u E_k = \lambda_k(u) E_k, \quad \lambda_k : U_N \longrightarrow \mu(K_k), \quad (3.20)$$

where $\mu(K_k)$ is the group of roots of unity of K_k and λ_k is a group homomorphism. Its conductor divides N . In particular:

- (1) if $K_k = \mathbb{Q}$ then λ_k is either trivial or a quadratic Dirichlet character modulo N ;
- (2) for every prime $p \nmid N$, the compression $A_k(p) = \Psi_p|_{V_k}$ satisfies

$$A_k(p) = \lambda_k(p) I_{V_k}, \quad \det(I - A_k(p)T) = \prod_{\iota} (1 - \iota(\lambda_k(p))T)^{d_k/[K_k:\mathbb{Q}]}, \quad (3.21)$$

the product running over the embeddings $\iota : K_k \hookrightarrow \mathbb{C}$ and $d_k = \dim_{\mathbb{Q}} V_k$. When $K_k = \mathbb{Q}$ this reduces to $\det(I - A_k(p)T) = (1 - \lambda_k(p)T)^{d_k}$.

Proof. By Corollary 2.21 the commutant A' is commutative and $\dim_{\mathbb{Q}} A' = \dim_{\mathbb{Q}} Z(A)$. Since $Z(A) \subseteq A'$ always, the two coincide, so $A' = Z(A) \subseteq A$.

Fix $u \in U_N$. By Proposition 3.2, $\Psi_u \in A' = Z(A)$. Multiplying by the primitive central idempotent E_k gives an element of $Z(AE_k) = K_k$, which is (3.20) once we note that K_k acts on the ideal AE_k through the scalar $\lambda_k(u)$. The map λ_k is multiplicative because $u \mapsto \Psi_u$ is (Lemma 3.1(4)) and E_k is idempotent. Since U_N is finite, $\lambda_k(u)$ is a root of unity in K_k . The conductor divides N because λ_k is defined on U_N and Ψ_n depends only on $n \bmod N$ by Lemma 3.1(1).

(1) The only roots of unity in $\mathbb{Q}$ are ± 1 , so a homomorphism $U_N \rightarrow \mu(\mathbb{Q}) = \{\pm 1\}$ is trivial or of order two; a homomorphism $U_N \rightarrow \{\pm 1\}$ is a real Dirichlet character modulo N , and a nontrivial one is quadratic.

(2) By (3.20), Ψ_p acts on $V_k = E_k V_G$ as multiplication by the scalar $\lambda_k(p) \in K_k$. The characteristic polynomial of multiplication by $\alpha \in K_k$ on a K_k -vector space of $\mathbb{Q}$ -dimension d_k is $\prod_{\iota} (T - \iota(\alpha))^{d_k/[K_k:\mathbb{Q}]}$, which gives (3.21). $\square$

3.7. Split multiplicity-one blocks.

Theorem 3.14. *Let $A = \mathcal{P}_{\text{ram}}(G)$, let E be a primitive central idempotent of A with center field $K = Z(AE)$. Assume that the simple factor is split over its center, $AE \cong M_r(K)$, and let m be the multiplicity of its natural simple module in V_G . If $m = 1$, then for every $u \in U_N$*

$$\Psi_u E = \lambda_E(u) E, \quad \lambda_E : U_N \longrightarrow \mu(K), \quad (3.22)$$

and λ_E is a homomorphism of conductor dividing N . No hypothesis is placed on the other blocks.

Proof. This is the rank-one specialization of Theorem 3.12. Splitness $AE \cong M_r(K)$ gives $D_j = K_j = K$ on the corresponding component, and multiplicity one gives $\dim_K M_j = 1$; by parts (3) and (4) the representation ρ_j is a scalar character $\lambda_E : U_N \rightarrow \mu(K)$, which is (3.22). The conductor divides N because Ψ_n depends only on $n \bmod N$ by Lemma 3.1(1). $\square$

For a general simple factor $AE \cong M_r(D)$ with center K , the double-centralizer theorem gives $EA'E \cong M_m(D^{\text{op}})$. For $m = 1$, the good-label action lies in D^{op} . If $D = K$, it is scalar.

3.8. The involution on the center.

Lemma 3.15. *Every primitive central idempotent E of $A = \mathcal{P}_{\text{ram}}(G)$ satisfies $E^* = E$, and the star restricts to an involution ι_E of the field $K = Z(AE)$.*

Proof. The class form (2.5) is positive definite and A is closed under its adjoint, so the isotypic decomposition of V_G under A is orthogonal and E is the orthogonal projection onto its isotypic component; an orthogonal projection is self-adjoint. The star is an anti-automorphism of A fixing E , so it preserves the ideal AE and hence its center K , on which an anti-automorphism is an automorphism; it is an involution because the star is. $\square$

Proposition 3.16. *With the hypotheses of Theorem 3.14,*

$$\lambda_E(u^{-1}) = \iota_E(\lambda_E(u)) = \lambda_E(u)^{-1} \quad (u \in U_N). \quad (3.23)$$

Consequently ι_E acts on the image group $\lambda_E(U_N)$ as inversion.

Proof. Apply the star to (3.22). On the left, $(\Psi_u E)^* = E^* \Psi_u^* = E \Psi_{u^{-1}} = \lambda_E(u^{-1}) E$ by Lemma 3.15 and (3.1). On the right, $\lambda_E(u)$ lies in $K \subseteq Z(A)$, so $(\lambda_E(u) E)^* = \iota_E(\lambda_E(u)) E$. Comparing gives the first equality in (3.23); the second is multiplicativity of λ_E . $\square$

Corollary 3.17. *Let $L_E = \mathbb{Q}(\lambda_E(U_N)) \subseteq K$ be the field generated by the realized values. The following alternatives are mutually exclusive and exhaustive.*

- (1) ι_E is trivial on L_E . Then $\lambda_E^2 = 1$, so λ_E is trivial or a quadratic Dirichlet character modulo N , and $L_E = \mathbb{Q}$.
- (2) ι_E is nontrivial on L_E . Then λ_E has order $e > 2$, $L_E = \mathbb{Q}(\zeta_e)$ is a CM field, and ι_E restricts to its complex conjugation.

In particular $K = \mathbb{Q}$ implies (1), but (1) is strictly weaker than $K = \mathbb{Q}$: what forces quadratic values is triviality of the star on the generated value field, not rationality of the center.

Proof. By (3.23), ι_E inverts every value. If ι_E is trivial on L_E then $\lambda_E(u) = \lambda_E(u)^{-1}$ for all u , so every value is ± 1 and $L_E = \mathbb{Q}$; Theorem 3.13(1) then identifies λ_E . Otherwise the image is cyclic of some order e , generated by a primitive e th root of unity ζ with $L_E = \mathbb{Q}(\zeta_e)$, and $\iota_E(\zeta) = \zeta^{-1}$ is a nontrivial automorphism, so $e > 2$. A nontrivial automorphism of $\mathbb{Q}(\zeta_e)$ inverting ζ_e is complex conjugation in every embedding, and a cyclotomic field of degree greater than one on which complex conjugation is nontrivial is CM. $\square$

A block carries a character of order greater than two only if its center admits a complex conjugation, and the values then generate a cyclotomic subfield of that center. The center fields occurring in Section 4.2 are accordingly $\mathbb{Q}$, $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$, and $\mathbb{Q}(\zeta_5)$; no real quadratic or real quartic center carries a nontrivial character.

3.9. Support and character order.

Lemma 3.18. *Let E be a central idempotent of $A = \mathcal{P}_{\text{ram}}(G)$ with $V_E = EV_G \neq 0$ on which the good labels act by scalars, $\Psi_u E = \lambda_E(u)E$. Let*

$$S(E) = \{C \in \text{Conj}(G) : f(C) \neq 0 \text{ for some } f \in V_E\}$$

be the class support of the block. Then $S(E)$ is stable under every σ_u , and for each $u \in U_N$ the multiplicative order of $\lambda_E(u)$ divides the order of the permutation $\sigma_u|_{S(E)}$. Consequently the order of λ_E divides the exponent of the image of U_N in the symmetric group of $S(E)$. In particular, if every $\Psi(U_N)$ -orbit meeting $S(E)$ has at most two classes, then λ_E is trivial or quadratic.

Proof. For $f \in V_E$ we have $\Psi_u f = \lambda_E(u)f$ up to the idempotent: $\Psi_u(Ef) = \lambda_E(u)Ef$, and $\lambda_E(u) \neq 0$, so $\Psi_u V_E = V_E$. Since $(\Psi_u f)(C) = f(\sigma_u C)$, the support satisfies $\sigma_u^{-1}S(E) \subseteq S(E)$, hence equality by finiteness. Now let m be the order of $\sigma_u|_{S(E)}$ and take $f \in V_E$. For $C \in S(E)$, $(\Psi_u^m f)(C) = f(\sigma_u^m C) = f(C)$; for $C \notin S(E)$, both sides vanish because $S(E)$ is σ_u -stable and f is supported on $S(E)$. Hence $\Psi_u^m f = f$, so $\lambda_E(u)^m = 1$. The exponent statement follows by taking least common multiples over u , and the final claim holds because a transitive abelian action is regular: on an orbit of at most two classes every σ_u squares to the identity, so $\lambda_E(u)^2 = 1$ for every u . $\square$

A character of order greater than two therefore requires a $\Psi(U_N)$ -orbit of at least three classes inside the block support.

Proposition 3.19. *Let $n \geq 1$.*

- (1) *Every conjugacy class of S_n is rational. Hence $\Psi_u = I$ for every $u \in U_N$, every representation ρ_j of Theorem 3.12 is trivial, and every good-label character of $\mathcal{P}_{\text{ram}}(S_n)$ is trivial.*
- (2) *Every $\Psi(U_N)$ -orbit on $\text{Conj}(A_n)$ has at most two classes. Hence every ρ_j has an elementary abelian 2-group as image, and every scalar block character of $\mathcal{P}_{\text{ram}}(A_n)$ is trivial or quadratic.*

Proof. (1) A permutation and its coprime powers have the same cycle type, hence are conjugate in S_n . So $\sigma_u = \text{id}$ for every good label u , and $\Psi_u = I$.

(2) Let $x \in A_n$ and let u be a good label. Then x^u has the same cycle type as x , so the A_n -class of x^u lies inside the S_n -class of x . An S_n -class meets A_n in at most two A_n -classes, so the $\Psi(U_N)$ -orbit of $[x]$ has at most two elements. Every σ_u therefore has order at most two on $\text{Conj}(A_n)$, so $\Psi_u^2 = I$, which forces $\rho_j(u)^2 = 1$ on every multiplicity space; the scalar statement is Lemma 3.18. $\square$

3.10. A trace identity. The following is the all-scalar case of (3.19).

Proposition 3.20. *Let $E_1, \dots, E_t$ be the primitive central idempotents of $A = \mathcal{P}_{\text{ram}}(G)$, with center fields K_k and $V_k = E_k V_G$, and suppose each acts by a center scalar, $\Psi_u E_k = \lambda_k(u)E_k$, for instance because every simple factor is split over its center and every block has multiplicity one (Theorem 3.14). Then for every $u \in U_N$*

$$\#\{C \in \text{Conj}(G) : \sigma_u(C) = C\} = \sum_{k=1}^t \frac{d_k}{[K_k : \mathbb{Q}]} \text{Tr}_{K_k/\mathbb{Q}}(\lambda_k(u)), \quad d_k = \dim_{\mathbb{Q}} V_k. \quad (3.24)$$

Proof. By (2.6) the operator Ψ_u is the permutation matrix of σ_u in the coordinate basis $(\delta_C)_C$, so its trace is the number of fixed classes. Trace is additive over $V_G = \bigoplus_k V_k$, and by hypothesis the restriction $\Psi_u|_{V_k}$ is multiplication by $\lambda_k(u) \in K_k$ on a K_k -vector space of $\mathbb{Q}$ -dimension d_k . The $\mathbb{Q}$ -trace of multiplication by $\alpha \in K_k$ on such a space is $(d_k/[K_k : \mathbb{Q}]) \text{Tr}_{K_k/\mathbb{Q}}(\alpha)$. $\square$

For a rational center, Theorem 3.13(1) gives a trivial or quadratic character. Under the isomorphism $U_N \cong \text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$, a quadratic character λ_k corresponds to a quadratic subfield $\mathbb{Q}(\sqrt{D_k}) \subseteq \mathbb{Q}(\zeta_N)$, and λ_k is the Kronecker symbol $(\frac{D_k}{\cdot})$ of the associated fundamental discriminant D_k , whose prime divisors divide N [8].

3.11. Saturation for finite groups.

Theorem 3.21. *If V_G is multiplicity free over $\mathcal{P}_{\text{ram}}(G)$, then*

$$\mathcal{P}_{\text{ram}}(G) = \mathcal{P}_{\text{Ad}}(G). \quad (3.25)$$

Proof. By Theorem 3.13 we have $A' = Z(A)$ for $A = \mathcal{P}_{\text{ram}}(G)$, and $\Psi_u \in A' = Z(A) \subseteq A$ for every $u \in U_N$. Hence $\Psi(U_N) \subseteq \mathcal{P}_{\text{ram}}(G)$, and (3.3) gives $\mathcal{P}_{\text{Ad}}(G) \subseteq \mathcal{P}_{\text{ram}}(G)$. The reverse inclusion is trivial. $\square$

3.12. The rational-class splitting.

Write $\Gamma = \Psi(U_N) \subseteq O(V_G)$ and

$$V_G^\Gamma = \{f \in V_G : \Psi_u f = f \text{ for all } u \in U_N\}.$$

Proposition 3.22. *The subspace V_G^Γ and its class-orthogonal complement are both reducing for $\mathcal{P}_{\text{Ad}}(G)$, and*

$$\dim_{\mathbb{Q}} V_G^\Gamma = \#\{\text{rational classes of } G\}. \quad (3.26)$$

If moreover V_G is multiplicity free over $A = \mathcal{P}_{\text{ram}}(G)$, then

$$V_G^\Gamma = \bigoplus_{\lambda_k=1} V_k, \quad \text{so} \quad \sum_{\lambda_k=1} d_k = \#\{\text{rational classes}\}, \quad \sum_{\lambda_k \neq 1} d_k = k(G) - \#\{\text{rational classes}\}. \quad (3.27)$$

Proof. Every element of Γ commutes with $\mathcal{P}_{\text{ram}}(G)$ by Proposition 3.2 and with Γ itself, so V_G^Γ is invariant under all of $\mathcal{P}_{\text{ram}}(G)$ and all of Γ ; by (3.3) it is invariant under $\mathcal{P}_{\text{Ad}}(G)$. Since Γ consists of isometries, the orthogonal complement of V_G^Γ is Γ -invariant, and it is $\mathcal{P}_{\text{Ad}}(G)$ -invariant because $\mathcal{P}_{\text{Ad}}(G)$ is adjoint closed and preserves V_G^Γ .

For (3.26): Γ acts on V_G by permuting the coordinate basis $(\delta_C)_C$ according to the σ_u , so the fixed space has as basis the orbit sums, one for each Γ -orbit on $\text{Conj}(G)$. By Lemma 3.1(2) and the remark following it, those orbits are the rational classes.

For (3.27): assume multiplicity freeness and fix $u \in U_N$. By (3.20), Ψ_u acts on V_k as multiplication by $\lambda_k(u) \in K_k$. If $\lambda_k = 1$ then $V_k \subseteq V_G^\Gamma$. If $\lambda_k \neq 1$, choose u with $\lambda_k(u) \neq 1$; then $\lambda_k(u) - 1$ is a nonzero element of the field K_k , hence invertible, so $\Psi_u - I$ is invertible on the K_k -module V_k and $V_k \cap V_G^\Gamma = 0$. As $V_G = \bigoplus_k V_k$ and each V_k is Γ -invariant, the fixed space is the sum of the trivial blocks. The two displayed sums then follow from $\sum_k d_k = k(G)$. $\square$

Proposition 3.22 is the general form of the coarse decomposition that any concrete census exhibits: the total class count splits as (rational classes) plus (nontrivially twisted dimension), and in the multiplicity-free case the splitting is by blocks. For the Monster, where $k(G) = 194$ and there are 172 rational classes, it forces $172 = \sum_{\lambda_k=1} d_k$ and $22 = \sum_{\lambda_k \neq 1} d_k$, and Proposition 4.17 identifies the blocks on either side.

Proposition 3.23. *Assume that V_G is multiplicity free over $\mathcal{P}_{\text{ram}}(G)$. Then every nontrivial good-label character λ_k is odd if and only if the number of real conjugacy classes of G equals the number of rational classes of G . Namely,*

$$\#\{\text{real classes}\} - \#\{\text{rational classes}\} = \sum_{\substack{\lambda_k \neq 1 \\ \lambda_k(-1)=1}} d_k. \quad (3.28)$$

Proof. The real classes are the fixed points of inversion on $\text{Conj}(G)$, so their number is the dimension of the $+1$ -eigenspace of Ψ_{-1} . By multiplicity freeness, Ψ_{-1} acts on V_k as the scalar $\lambda_k(-1)$, and hence

$$\#\{\text{real classes}\} = \sum_{\lambda_k(-1)=1} d_k.$$

By (3.27), the sum over the trivial characters is the number of rational classes. Subtracting gives (3.28). The right-hand side vanishes if and only if every nontrivial character is odd. $\square$

3.13. Compressions. Channel compressions obey the multiplicative law needed to define local block data without an ordering convention.

Proposition 3.24. *Let E be a central idempotent of $A = \mathcal{P}_{\text{ram}}(G)$, let $V_E = EV_G$, and for $n \geq 1$ put $A_E(n) = E\Psi_n E|_{V_E}$. Then E commutes with Ψ_n for every $n \geq 1$, and*

$$A_E(m)A_E(n) = A_E(mn) = A_E(n)A_E(m) \quad (m, n \geq 1). \quad (3.29)$$

Proof. For a ramified label n we have $\Psi_n \in A$ and $E \in Z(A)$, so $E\Psi_n = \Psi_n E$. For a good label n we have $\Psi_n \in A'$ by Proposition 3.2 and $E \in A$, so again $E\Psi_n = \Psi_n E$. A general n is a product of the two kinds by Proposition 3.3. Therefore

$$E\Psi_m E \cdot E\Psi_n E = E^3 \Psi_m \Psi_n = E\Psi_{mn},$$

and restricting to V_E gives (3.29); the symmetry in m, n holds because $\Psi_m \Psi_n = \Psi_{mn} = \Psi_n \Psi_m$. $\square$

4. THE SPORADIC SIMPLE GROUPS

4.1. Rational factors for the sporadic simple groups. Names and class counts are those of the *Atlas* [6]. Write

$$K_{-3} = \mathbb{Q}(\sqrt{-3}), \quad K_5 = \mathbb{Q}(\zeta_5).$$

The degree of $K_5/\mathbb{Q}$ is four. Hence a factor $M_r(K_5)$ gives four conjugate copies of $M_r(\mathbb{C})$ after scalar extension.

Lemma 4.1. *Let $B = M_r(D)$ be a simple rational factor with center K , Schur index $s = \sqrt{\dim_K D}$, and split degree $d = rs$. Suppose the associated B -module in rational class-coordinate space has complex multiplicity m . Then s divides both d and m . In particular, $\gcd(d, m) = 1$ forces $D = K$. For the possible indices in the rational group-algebra setting see [24].*

Proof. For each embedding $K \hookrightarrow \mathbb{C}$, scalar extension sends B to $M_{rs}(\mathbb{C})$. A simple left B -module has K -dimension rs^2 and complexifies to s copies of the natural rs -dimensional module [18, §29]. Every rational B -module therefore gives a complex multiplicity divisible by s , while s also divides $d = rs$. $\square$

Theorem 4.2. *The rational algebras of the twenty-six sporadic simple groups are listed below. The columns z and c give $\dim_{\mathbb{Q}} Z(A)$ and $\dim_{\mathbb{Q}} A'$. Every simple factor has Schur index one. The only center fields are $\mathbb{Q}$, K_{-3} , and K_5 , and K_5 occurs only for Ly .*

G	$k(G)$	$\dim A$	z	c	A over $\mathbb{Q}$
M_{11}	10	66	3	3	$M_8(\mathbb{Q}) \oplus \mathbb{Q}^2$
M_{12}	15	149	3	3	$M_{12}(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q}$
M_{22}	12	102	3	3	$M_{10}(\mathbb{Q}) \oplus \mathbb{Q}^2$
M_{23}	17	151	5	5	$M_{12}(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q}^3$
M_{24}	26	452	4	4	$M_{21}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus \mathbb{Q}^2$
J_1	15	111	4	4	$M_{10}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus K_{-3}$
J_2	21	269	3	3	$M_{16}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q})$
J_3	21	209	6	6	$M_{14}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus \mathbb{Q}^2 \oplus K_{-3}$
J_4	62	2000	11	11	$M_{44}(\mathbb{Q}) \oplus M_7(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q} \oplus K_{-3}^3$
HS	24	444	4	4	$M_{21}(\mathbb{Q}) \oplus \mathbb{Q}^3$
McL	24	334	5	5	$M_{18}(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q}^2$
He	33	647	5	5	$M_{25}(\mathbb{Q}) \oplus M_4(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q}^2$
Ru	36	792	9	9	$M_{28}(\mathbb{Q}) \oplus \mathbb{Q}^4 \oplus K_{-3}^2$
Suz	43	1381	5	5	$M_{37}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus \mathbb{Q}^3$
$O'N$	30	411	9	12	$M_{20}(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q}^5 \oplus K_{-3}$
Co_1	101	9415	4	4	$M_{97}(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q}^2$
Co_2	60	2926	5	5	$M_{54}(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q}^2$

G	$k(G)$	$\dim A$	z	c	A over $\mathbb{Q}$
Co_3	42	1306	5	5	$M_{36}(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q}^2$
Fi_{22}	65	3261	7	7	$M_{57}(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q}^4$
Fi_{23}	98	8294	6	6	$M_{91}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus \mathbb{Q}^4$
Fi'_{24}	108	8853	12	15	$M_{94}(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q}^9$
HN	54	1879	7	10	$M_{43}(\mathbb{Q}) \oplus M_5(\mathbb{Q}) \oplus \mathbb{Q}^5$
Ly	53	1543	12	12	$M_{39}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q}^3 \oplus K_{-3} \oplus K_5$
Th	48	1384	10	10	$M_{37}(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q}^7$
$\mathbb{B}$	184	26598	13	19	$M_{163}(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q})^3 \oplus \mathbb{Q}^8$
$\mathbb{M}$	194	28958	15	15	$M_{170}(\mathbb{Q}) \oplus M_5(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q})^4 \oplus \mathbb{Q}^8$

Proof. The computation described in [Section 6](#) constructs each rational word algebra and verifies closure. A separating central element gives square-free center factors; rational central idempotents give the dimensions of the corresponding rational ideals. These data determine the split degree and class-space multiplicity of each complex block. Their gcd is one, so [Lemma 4.1](#) gives Schur index one.

The quadratic defining polynomials have square-free discriminant -3 . The only quartic center is defined by $x^4 + x^3 + x^2 + x + 1$, which gives K_5 . Ranks at the primes 1,000,003 and 1,000,033 reproduce the algebra, center, and commutant dimensions of each residual calculation. $\square$

Corollary 4.3. *For each of the twenty-two sporadic groups whose class-coordinate representation is multiplicity free, $\mathcal{P}_{\text{ram}}(G) = \mathcal{P}_{\text{Ad}}(G)$. In particular the Monster satisfies $\mathcal{P}_{\text{ram}}(\mathbb{M}) = \mathcal{P}_{\text{Ad}}(\mathbb{M})$.*

Proof. Immediate from [Theorem 3.21](#) and the equality $z = c$ recorded in the table. $\square$

The class-coordinate representation is multiplicity-free for 22 sporadic groups. The exceptions are $O'N$, Fi'_{24} , HN , and $\mathbb{B}$. For the first three, a scalar complex block occurs with multiplicity two. For $\mathbb{B}$, one scalar block and one 3-dimensional block each occur with multiplicity two. The distinction between z and c in the table makes these multiplicities visible.

4.2. Good-label characters. The sporadic census contains nine blocks with center $\mathbb{Q}(\zeta_3)$ and one block with center $\mathbb{Q}(\zeta_5)$.

Throughout, $N = \exp(G)$, $U_N = (\mathbb{Z}/N)^\times$, $\Gamma = \Psi(U_N)$, and χ_D denotes the Kronecker symbol $(\frac{D}{\cdot})$ of a fundamental discriminant D .

4.2.1. Recovering the unit action from ramified data. The CTblLib input consists of class sizes, element orders, and the power maps σ_p for primes $p \mid N$.

Definition 4.4. *For an integer $m \mid N$ put*

$$H_m = \langle p \bmod m : p \text{ prime}, p \mid N, p \nmid m \rangle \leq U_m. \quad (4.1)$$

We call G ramified transparent if $H_m = U_m$ for every element order m of G .

Theorem 4.5. *Let C be a conjugacy class whose elements have order m , let $u \in U_N$, and let v be any ramified label with $v \equiv u \pmod{m}$. Then*

$$\sigma_u(C) = \sigma_v(C). \quad (4.2)$$

If G is ramified transparent, the maps $\{\sigma_p : p \mid N\}$ determine the action of U_N on $\text{Conj}(G)$.

Proof. For $x \in C$ we have $x^m = 1$, so $x^u = x^{u \bmod m} = x^{v \bmod m} = x^v$, whence $\sigma_u(C) = \sigma_v(C)$, which is (4.2). A ramified v with $v \equiv u \pmod{m}$ exists if and only if $u \bmod m \in H_m$, because the residues of ramified labels modulo m that are units are the elements of H_m : a ramified v is a product of primes dividing N , those dividing m contribute non-units, and $\gcd(v, m) = 1$ forces every prime factor of v to be coprime to m . Under ramified transparency this holds for all m and u , and σ_v is computed from the stored maps by multiplicativity, $\sigma_v = \prod_p \sigma_p^{e_p}$ for $v = \prod_p p^{e_p}$. $\square$

Proposition 4.6. *Twenty-four of the twenty-six sporadic simple groups are ramified transparent. The exceptions are J_3 , where $[U_{15} : H_{15}] = 2$, and Th , where $[U_{30} : H_{30}] = 2$; in both cases the missing coset is the one containing -1 , and the ambiguity concerns a single pair of classes.*

Proof. Closing the residues of the primes dividing N under multiplication modulo m and comparing $|H_m|$ with $\varphi(m)$ leaves only the two displayed failures. For J_3 , $N = 2^3 \cdot 3^2 \cdot 5 \cdot 17 \cdot 19$ and the primes coprime to 15 reduce to $2, 17, 19 \equiv 2, 2, 4 \pmod{15}$, generating $\langle 2 \rangle = \{1, 2, 4, 8\}$ of index two in U_{15} ; the missing coset is $7\langle 2 \rangle = \{7, 11, 13, 14\} \ni -1$. The computation for Th at $m = 30$ is identical in form. $\square$

For J_3 and Th , call a permutation of $\text{Conj}(G)$ *admissible* if it extends the forced values, preserves class size and element order, and commutes with the stored power maps.

Proposition 4.7. *For each sporadic simple group and each chosen generator of U_N , the admissible completion is unique. The stored ramified maps, class sizes, and element orders determine the action of U_N on $\text{Conj}(G)$.*

Proof. For the twenty-four ramified transparent groups, Theorem 4.5 determines all images. For J_3 , two of the six chosen generators, those with nontrivial 3- and 5-components, leave undetermined the two classes of element order 15, the remaining nineteen images being forced; of the two possible assignments only one commutes with σ_3 . For Th , one of the eight generators, the 3-component, leaves the two classes of element order 30 undetermined out of forty-eight, and there the disambiguating constraint is σ_2 . Which of the two assignments survives is not uniform: the surviving completion fixes both classes for the 3-component of J_3 and transposes them in the other two cases. $\square$

The statement is independent of the displayed generators: once the permutations are recovered on any generating set, the homomorphism $U_N \rightarrow \text{Sym}(\text{Conj}(G))$ determines every other choice.

4.2.2. The character census.

Theorem 4.8. *Let G be a sporadic simple group and E_k a primitive central idempotent of $\mathcal{P}_{\text{ram}}(G)$. There is a character $\lambda_k : U_N \rightarrow \mu(K_k)$ with*

$$\Psi_u E_k = \lambda_k(u) E_k.$$

The principal block carries the trivial character. Of the remaining blocks, ten have nonrational center and are listed in Section 4.2.3; every other one carries the trivial character or the Kronecker symbol χ_{D_k} of a fundamental discriminant D_k whose prime divisors divide N . Five blocks, in $O'N$, Fi'_{24} , HN , and $\mathbb{B}$, have multiplicity two, and Ψ_u is scalar on those as well.

Proof. Compute the primitive central idempotents over $\mathbb{Q}$ and the operators Ψ_u for one generator per cyclic factor of $\prod_{p^a \parallel N} U_{p^a}$. The block-local theorem treats the multiplicity-one blocks. Rational row reduction gives $\Psi_u E_k = \lambda_k(u) E_k$ on the five multiplicity-two blocks. The order of $\lambda_k(u)$ is the least e with $(\Psi_u E_k)^e = E_k$; the character order is the least common multiple over the generators; the conductor is assembled from the local orders by the standard rule for U_{p^a} . For a rational block the parity $\lambda_k(-1)$ selects the sign of the fundamental discriminant of the given conductor. Direct substitution gives $\lambda_k(u) = \chi_D(u)$ on the generators.

The principal block character is trivial. That block is the cyclic module $A\delta_{1A}$ generated by the identity-class coordinate (Section 6); every Ψ_u fixes δ_{1A} and commutes with A , so it fixes $A\delta_{1A}$ pointwise. $\square$

4.2.3. The ten nonrational blocks. The ten nonrational blocks occur in

$$J_1, \quad J_3, \quad J_4, \quad Ru, \quad O'N, \quad Ly.$$

Nine blocks have center $\mathbb{Q}(\zeta_3)$; the unique remaining block has center $\mathbb{Q}(\zeta_5)$ and belongs to Ly . In every case the star acts as complex conjugation on the center, the character order is the odd prime

e with $K = \mathbb{Q}(\zeta_e)$, and the block is supported on e classes of a single element order. Its split degree is $d_k = \varphi(e) = e - 1$. The complete list is:

G	K_k	e	cond.	elt. order
J_1	K_{-3}	3	19	19
J_3	K_{-3}	3	9	9
J_4	K_{-3}	3	37	37
J_4	K_{-3}	3	31	31
J_4	K_{-3}	3	43	43
Ru	K_{-3}	3	7	14
Ru	K_{-3}	3	13	26
$O'N$	K_{-3}	3	19	19
Ly	K_{-3}	3	67	67
Ly	K_5	5	31	31

Remark 4.9. The six groups carrying a nonrational center are exactly the six pariah groups, the sporadic groups not involved in the Monster. Every group of the happy family has $\mathcal{P}_{\text{ram}}(G)$ split over $\mathbb{Q}$ with all centers rational. We record this as an observation of the census; no explanation is offered here.

Proposition 4.10. *Let E be a block whose support consists of classes of a common element order m , and let λ_E be its good-label character, of order $e > 1$. Write $m = 2^a m'$ with m' odd. Then*

- (1) λ_E factors through U_m , so its conductor divides m ;
- (2) if $a \leq 1$, the conductor divides m' ;
- (3) if moreover $m' = p^b$ is a prime power and $e \nmid \varphi(p^{b-1})$, the conductor equals m' .

Proof. (1) By Theorem 4.5 the permutation σ_u acts on a class of element order m through $u \bmod m$ alone, so λ_E factors through U_m and its conductor divides m .

(2) For odd m' the Chinese remainder theorem gives $U_{2m'} \cong U_2 \times U_{m'} = U_{m'}$, so when $a \leq 1$ the character factors through $U_{m'}$.

(3) The conductor is p^c for some $c \leq b$, minimal such that λ_E factors through U_{p^c} ; then $e = |\lambda_E(U_N)|$ divides $|U_{p^c}| = \varphi(p^c)$. For p odd and $c \leq b - 1$ one has $\varphi(p^c) \mid \varphi(p^{b-1})$, whence $e \mid \varphi(p^{b-1})$, contrary to hypothesis. Hence $c = b$. $\square$

Corollary 4.11. *In every row of Section 4.2.3 the conductor equals the odd part m' of the common element order.*

Proof. Each row has $a \leq 1$, and m' is a prime power. In nine rows m' is one of 7, 13, 19, 31, 37, 43, 67; then $b = 1$, and $e > 1$ does not divide $\varphi(p^0) = 1$. In the remaining row $m' = 9$ and $e = 3$, which does not divide $\varphi(3) = 2$. Proposition 4.10 applies in every case. $\square$

Example 4.12. *Among the twenty-six sporadic groups, Ly alone has a quartic center. Its five conjugacy classes of elements of order 31 form one rational class and carry the full $\mathbb{Q}(\zeta_5)$ factor of $\mathcal{P}_{\text{ram}}(Ly)$. Writing R for this five-class orbit, the block is*

$$V_k = \left\{ f \in \text{span}_{\mathbb{Q}}(\delta_C : C \in R) : \sum_{C \in R} f(C) = 0 \right\}, \quad \dim_{\mathbb{Q}} V_k = 4,$$

and the good labels act through a quintic character of conductor 31. Thus V_k is one-dimensional over its cyclotomic endomorphism field:

$$\text{End}_{\mathbb{Q}[U_N]}(V_k) = \mathbb{Q}(\zeta_5).$$

The character is even, so inversion fixes all five order-31 classes.

Example 4.13. *The three classes of elements of order 19 in J_1 form a single rational class; the augmentation-zero part of their span is a two-dimensional block with center $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$, on which the good labels act through a cubic character of conductor 19. Since $19 \equiv 1 \pmod{3}$, such characters exist modulo 19; there are two of them, complex conjugate to one another, and they are interchanged by the choice of embedding $\mathbb{Q}(\zeta_3) \hookrightarrow \mathbb{C}$. The pair, not either member, is the invariant.*

Theorem 4.14. *Let $R \subseteq \text{Conj}(G)$ be a Γ -orbit with $|R| = r$, let*

$$W_R = \left\{ f \in \text{span}_{\mathbb{Q}}(\delta_C : C \in R) : \sum_{C \in R} f(C) = 0 \right\}, \quad \dim_{\mathbb{Q}} W_R = r - 1.$$

Then Γ acts regularly on R through an abelian quotient A_R of order r , and W_R is Γ -stable. If r is prime, then $A_R \cong \mathbb{Z}/r$, and Γ acts on W_R through a character of order r with values in $\mathbb{Q}(\zeta_r)$, and $\text{End}_{\mathbb{Q}[\Gamma]}(W_R) = \mathbb{Q}(\zeta_r)$. If moreover W_R is a single split block of $\mathcal{P}_{\text{ram}}(G)$ of multiplicity one, that block has center $K = \mathbb{Q}(\zeta_r)$ and good-label character of order r .

Proof. Γ is abelian by Lemma 3.1(4) and acts transitively on R by definition of an orbit. In a transitive action of an abelian group all point stabilizers agree; after quotienting by their common value Γ_R , the action is free and transitive. Thus $A_R = \Gamma/\Gamma_R$ has order r , and R is an A_R -torsor. If r is prime then $A_R \cong \mathbb{Z}/r$. The span of R is the regular representation $\mathbb{Q}[A_R]$, which splits as $\mathbb{Q} \oplus W_R$ with $\mathbb{Q}$ the invariants; both summands are Γ -stable.

Suppose r is prime. Then $\mathbb{Q}[\mathbb{Z}/r] \cong \mathbb{Q} \times \mathbb{Q}(\zeta_r)$ [22], the second factor acting on W_R , so W_R is a one-dimensional $\mathbb{Q}(\zeta_r)$ -vector space on which a generator of $\mathbb{Z}/r$ acts as multiplication by ζ_r . Hence the induced character has order r , its values generate $\mathbb{Q}(\zeta_r)$, and the commuting algebra is $\mathbb{Q}(\zeta_r)$. If W_R is a single split multiplicity-one block of $\mathcal{P}_{\text{ram}}(G)$, its center is its commuting algebra inside the isotypic component, which contains and is contained in $\mathbb{Q}(\zeta_r)$ by the previous sentence and Theorem 3.14. $\square$

Corollary 4.15. *Let r be an odd prime. Then V_G has a $\mathbb{Q}[U_N]$ -summand with endomorphism field $\mathbb{Q}(\zeta_r)$ if and only if r divides $|R|$ for some rational class R .*

Proof. By Theorem 3.4, $V_G \cong \bigoplus_R \mathbb{Q}[A_R]$ with $A_R = U_N/\Gamma_R$ abelian of order $|R|$. For an abelian group A , the simple summands of $\mathbb{Q}[A]$ are the fields $\mathbb{Q}(\chi) = \mathbb{Q}(\zeta_{\text{ord}\chi})$ as χ runs over the Galois orbits of characters of A , each acting on itself, so the endomorphism field of that summand is $\mathbb{Q}(\chi)$. Since r is an odd prime, $\mathbb{Q}(\chi) = \mathbb{Q}(\zeta_r)$ holds if and only if $\text{ord}\chi$ is r or $2r$, and such a χ exists for some A_R if and only if r divides the exponent of some A_R , equivalently $r \mid |A_R| = |R|$. $\square$

Corollary 4.16. *Every nonrational block in the sporadic census arises as in Theorem 4.14: nine from rational classes of size three, giving $K = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ and cubic characters, and one, the $\mathbb{Q}(\zeta_5)$ block of Ly , from a rational class of size five, giving a quintic character. In every case $d_k = r - 1$.*

Proof. For each block, compute its class support. For all ten nonrational blocks the support is a single orbit R of prime size $r \in \{3, 5\}$, all of whose classes have one element order, and $d_k = |R| - 1$; the block therefore is W_R and Theorem 4.14 applies. Comparing its conclusion with the computed columns, $K_k = \mathbb{Q}(\zeta_r)$ and the character order is r , as tabulated. $\square$

Theorem 4.14 with $r = 2$ accounts for a large part of the rational half of Theorem 4.8 as well: a rational class splitting into two conjugacy classes contributes a one-dimensional block on which Γ acts by a quadratic character. The blocks with $d_k > r - 1$ are the ones where several rational classes fuse into a single simple factor; those blocks have a conductor divisible by more than one prime.

4.2.4. The Monster.

Proposition 4.17. *Let $\mathbb{M}$ be the Monster, so that $k(\mathbb{M}) = 194$ and*

$$N = \exp(\mathbb{M}) = 2^5 3^3 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71. \quad (4.3)$$

Its ramified algebra is

$$\mathcal{P}_{\text{ram}}(\mathbb{M}) \cong M_{170}(\mathbb{Q}) \oplus M_5(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q})^4 \oplus \mathbb{Q}^8.$$

Thus all fifteen blocks have center $\mathbb{Q}$. Two carry the trivial character; the other thirteen carry quadratic characters whose fundamental discriminants are

$$-11, -20, -23, -31, -39, -47, -56, -59, -71, -87, -95, -104, -119. \quad (4.4)$$

All thirteen are negative, so every nontrivial good-label character of the Monster is odd: $\lambda_k(-1) = -1$ on each of the thirteen blocks.

Proof. The rational decomposition and the character conductors are the Monster row of Theorems 4.2 and 4.8. The inversion map in the CTblLib class ordering fixes 172 of the 194 classes, while the unit-group action has 172 orbits. Hence the numbers of real and rational classes agree. The defining module is multiplicity free, so Proposition 3.23 forces every nontrivial block character to be odd. The thirteen fundamental discriminants are then determined by their certified conductors and their values on the unit-group generators. $\square$

By Proposition 3.22, the $+1$ eigenspace of Ψ_{-1} is spanned by the orbit sums of the inversion action, so (4.4) says that every nontrivially twisted block of the Monster is detected already by inversion: the twenty-two dimensions $194 - 172$ lie entirely in the -1 eigenspace of complex conjugation on classes. This is special to $\mathbb{M}$ among the large sporadic groups: Ly , for instance, has four blocks with positive discriminant 40, 37, 24, 21 and one with -11 , and J_4 has three positive and one negative.

The assignment

$$G \longmapsto \{(K_k, d_k, \lambda_k)\}_k$$

is an invariant of the marked weighted power system. The groups Co_2 and Co_3 have the same rational block list $M_r(\mathbb{Q}) \oplus M_2(\mathbb{Q})^2 \oplus \mathbb{Q}^2$ up to the principal degree, but their residual characters differ, χ_{-23}, χ_{-7} and χ_{-20}, χ_{-23} , respectively.

4.3. Rank-one saturation. Let a finite elementary abelian 2-group K act by weight-preserving permutations on a finite weighted set X . Write

$$V_X = \bigoplus_{\chi \in \widehat{K}} V_\chi, \quad V_\chi = \{v : kv = \chi(k)v \text{ for all } k \in K\}.$$

Every V_χ is rational. Hence

$$\text{End}_{\mathbb{Q}[K]}(V_X) = \bigoplus_{\chi: m_\chi > 0} M_{m_\chi}(\mathbb{Q}), \quad m_\chi = \dim_{\mathbb{Q}} V_\chi.$$

Lemma 4.18 (Cyclic rank-one generation). *Let W be a finite-dimensional rational inner-product space, let $B \subseteq \text{End}_{\mathbb{Q}}(W)$ be a unital star-algebra, and let $v \in W$. If the orthogonal rank-one projection E_v belongs to B and $Bv = W$, then $B = \text{End}_{\mathbb{Q}}(W)$.*

Proof. Choose $b_1, \dots, b_d \in B$ such that $b_1v, \dots, b_dv$ is a basis of W . The operators $b_i E_v b_j^*$ are nonzero scalar multiples of the matrix units for this basis. They span $\text{End}_{\mathbb{Q}}(W)$. $\square$

For a in the monoid generated by the labels, put

$$r_a = P_a^* P_a \mathbf{1}.$$

The transfer identity makes $P_a^* P_a = M_{r_a}$. A finite tuple of the functions r_a will be called a root profile.

Lemma 4.19 (Root-profile projectors). *Let $O \subseteq X$ be a K -orbit. If a finite root profile takes one joint value on O and a different joint value on every other orbit, then the coordinate projection E_O belongs to $\mathcal{P}_E(\mathcal{X})$.*

Proof. The diagonal operators M_{r_a} belong to $\mathcal{P}_E(\mathcal{X})$. Multivariate Lagrange interpolation in the finitely many joint values produces the indicator of O . $\square$

Suppose $O = \{x, y\}$ is a two-point orbit belonging to the character χ . Since K preserves the weight, $\mu(x) = \mu(y)$, and

$$E_O = E_{\delta_x + \delta_y} + E_{\delta_x - \delta_y}$$

after normalizing the two orthogonal vectors. Once the trivial matrix block has been generated, an isolated root profile for O therefore supplies the rank-one projection onto $\mathbb{Q}(\delta_x - \delta_y) \subseteq V_\chi$.

Theorem 4.20 (Root-cyclic saturation). *Let $A = \mathcal{P}_E(\mathcal{X}) \subseteq \text{End}_{\mathbb{Q}[K]}(V_X)$, where K is elementary abelian of exponent 2. Assume:*

- (1) *a fixed point x_0 has $E_{x_0} \in A$, and $A\delta_{x_0} = V_1$;*
- (2) *for each nontrivial χ with $m_\chi > 1$, a two-point orbit $O_\chi = \{x_\chi, y_\chi\}$ has an isolated root profile and $A(\delta_{x_\chi} - \delta_{y_\chi}) = V_\chi$;*
- (3) *the restrictions of elements of A separate the characters χ with $m_\chi = 1$.*

Then

$$A = \text{End}_{\mathbb{Q}[K]}(V_X).$$

The saturation certificate requires

$$m_1 + \sum_{\substack{\chi \neq 1 \\ m_\chi > 1}} m_\chi$$

orbit vectors, together with scalar eigenvalue data on the one-dimensional channels.

Proof. The first hypothesis and Lemma 4.18 generate the full trivial block. For every higher-dimensional nontrivial channel, Lemma 4.19 and subtraction of the trivial rank-one summand give the projection onto $\mathbb{Q}(\delta_{x_\chi} - \delta_{y_\chi})$. The cyclic rank-one lemma then generates $\text{End}_{\mathbb{Q}}(V_\chi)$. Project away these matrix blocks. On the remaining one-dimensional channels, the separating restrictions generate all coordinate idempotents by interpolation. Thus A contains each simple factor of the displayed commutant. $\square$

Lemma 4.21 (Modular cyclicity). *Let $R = \mathbb{Z}[1/M]$, let $B \subseteq M_d(R)$ be generated by fixed matrices, and let $v \in R^d$. Let $\ell \nmid M$ be prime. If words $w_1, \dots, w_d$ in the generators satisfy*

$$\det[w_1 v \ \cdots \ w_d v] \not\equiv 0 \pmod{\ell},$$

then v is cyclic over $\mathbb{Q}$.

Proof. The orbit determinant belongs to R . Its nonzero reduction cannot come from the zero rational number. $\square$

4.4. The Monster symmetry commutant. Let $\Gamma_{\mathbb{M}}$ be the group of permutations of the 194 Monster classes that preserve class size and commute with every prime-power map. It is the automorphism group of the marked weighted power system.

Theorem 4.22 (Weighted-power symmetry and commutant). *The group $\Gamma_{\mathbb{M}}$ is elementary abelian of rank 14. Its subgroup generated by the nineteen stored ordinary-table automorphisms has rank 13, and a complementary factor is generated by the involution*

$$(16B \ 16C)(32A \ 32B).$$

For the permutation representation on $V_{\mathbb{M}} = \mathbb{Q}^{\text{Conj}(\mathbb{M})}$,

$$\text{End}_{\mathbb{Q}\Gamma_{\mathbb{M}}}(V_{\mathbb{M}}) = \mathcal{P}_{\text{ram}}(\mathbb{M}) \cong M_{170}(\mathbb{Q}) \oplus M_5(\mathbb{Q}) \oplus M_3(\mathbb{Q}) \oplus M_2(\mathbb{Q})^{\oplus 4} \oplus \mathbb{Q}^{\oplus 8}.$$

Its dimension is 28958; its center and commutant have dimension 15.

Proof. Start with the coloring by class size and element order. Refine it by the colors of all prime-power images and by the weighted color counts in each reverse fibre. The first two refinement rounds have 161 and 170 cells, and the second coloring is stable. Its nonsingleton cells are twenty-four pairs. Commutation with the fifteen prime-power maps joins their swap variables into fourteen components of sizes

$$2, 5, 3, 2, 1, 2, 2, 1, 1, 1, 1, 1, 1, 1.$$

Every component swap satisfies all power-map equations. Hence $\Gamma_{\mathbb{M}} \cong (C_2)^{14}$. The binary matrix of the nineteen stored table automorphisms has rank 13; its missing coordinate is the component $(16B \ 16C)(32A \ 32B)$.

Each singleton contributes one trivial vector. Each pair contributes its sum to the trivial channel and its difference to the character of its swap component. Thus

$$V_{\mathbb{M}} \cong \mathbb{Q}_1^{170} \oplus \mathbb{Q}_{\chi_1}^2 \oplus \mathbb{Q}_{\chi_2}^5 \oplus \mathbb{Q}_{\chi_3}^3 \oplus \mathbb{Q}_{\chi_4}^2 \oplus \mathbb{Q}_{\chi_6}^2 \oplus \mathbb{Q}_{\chi_7}^2 \oplus \bigoplus_{i \in \{5,8,9,10,11,12,13,14\}} \mathbb{Q}_{\chi_i}.$$

This gives the displayed target commutant.

It remains to prove saturation. Since every Monster element has exponent dividing $N = \exp(\mathbb{M})$, the operator $|\mathbb{M}|^{-1} P_N^* P_N$ is the coordinate projection onto 1A. Modulo 1000003, the orbit of δ_{1A} has rank 170, with a retained basis of words of length at most six. The pairs

$$16B/C, 23A/B, 31A/B, 39C/D, 44A/B, 47A/B$$

have isolated root profiles for the six nontrivial channels of dimension greater than one. Their orbit determinants modulo 1000003 are

$$1, 2, 1, 1, -1, 1,$$

with maximum word lengths 1, 3, 1, 1, 1, 1. Finally, the restrictions of (P_2, P_5, P_7) to the eight scalar channels are pairwise distinct. The hypotheses of Theorem 4.20 hold, so $\mathcal{P}_{\text{ram}}(\mathbb{M}) = \text{End}_{\mathbb{Q}\Gamma_{\mathbb{M}}}(V_{\mathbb{M}})$. The certificate uses $170 + 2 + 5 + 3 + 2 + 2 + 2 = 186$ retained orbit vectors. All denominators are invertible modulo 1000003, so Lemma 4.21 lifts the seven orbit determinants to $\mathbb{Q}$. $\square$

Remark 4.23. The exceptional rational $M_2(\mathbb{Q})$ -channel is the unique symmetry direction on which the complementary involution acts and the stored table-automorphism subgroup acts trivially. No identification of the stored subgroup with the full unit-power image is used.

4.5. Symmetries invisible to the character table. Galois conjugation of classes is an automorphism of the character table and of the weighted power system, so $\Psi(U_N) \subseteq \Gamma_G$ for every G , where Γ_G is the automorphism group of the weighted power system. Theorem 4.22 shows that the inclusion can be strict: for the Monster, $\Gamma_{\mathbb{M}} \cong (C_2)^{14}$ while the Galois image has rank 13, the thirteen distinct quadratic characters of Proposition 4.17. The complementary involution $(16B \ 16C)(32A \ 32B)$ preserves class sizes and all fifteen prime-power maps. It lies outside the group generated by the stored automorphisms of the character table, so it is not induced by any automorphism of that table. Thus the classes $16B, 16C$ and $32A, 32B$ are distinguished by the character table but not by class sizes and power maps.

At the level of the algebra this involution is visible as the unique nonprincipal block of $\mathcal{P}_{\text{ram}}(\mathbb{M})$ on which the good labels act trivially: by Proposition 3.22 the Galois-fixed space $V_{\mathbb{M}}^{\Gamma}$ has dimension 172, and it decomposes under $\mathcal{P}_{\text{ram}}(\mathbb{M})$ as the 170-dimensional principal block plus one $M_2(\mathbb{Q})$ -block, the difference channel of the complementary involution.

The same signature occurs elsewhere in the census. In the character table of Theorem 4.8, Co_2 and Co_3 each have an $M_2(\mathbb{Q})$ -block carrying the trivial character, so each has a two-dimensional nonprincipal simple factor inside its Galois-fixed space. By Theorem 2.18 that factor comes from a balanced kernel commuting with $\Psi(U_N)$ and not accounted for by it. Whether it arises from a permutation of classes, as for the Monster, is not determined by the computations reported here; the symmetry group Γ_G was computed only for $\mathbb{M}$.

For Fi_{23} the answer is known and is negative. Its census row has a scalar block with trivial character spanned by

$$v = \delta_{12H} - \frac{1}{3} \delta_{12L} - \frac{1}{4} \delta_{12M} + \frac{1}{12} \delta_{12O},$$

a joint eigenvector of all sixteen generators, with eigenvalue 0 for $\Psi_2, \Psi_2^*, \Psi_3, \Psi_3^*$ and 1 for the others, orthogonal to the principal block $A\delta_{1A}$, which has exact dimension 91 (Section 6). The four coefficients of v have distinct absolute values, so no permutation of classes commuting with the power maps moves v except trivially; every element of $\mathbb{Q}\Gamma_{Fi_{23}}$ acts on the Galois-fixed space V^Γ as a scalar, while the projection onto $\mathbb{Q}v$ lies in the commutant and acts on V^Γ nontrivially. Hence $\text{Bal}(Fi_{23}) \supsetneq \mathbb{Q}\Gamma_{Fi_{23}}$: the commutant of $\mathcal{P}_{\text{ram}}(Fi_{23})$ contains a balanced kernel that is not a combination of permutation symmetries. The Monster's extra block is a permutation symmetry invisible to the character table; the Fi_{23} block is a fractional symmetry invisible even to the automorphism group of the weighted power system.

4.6. Computational record. The file `good_label_characters.json` records, per group, the generators of U_N and the reconstruction data of Theorem 4.5, and per primitive central idempotent the center field and its defining polynomial, d_k , the class support with its element orders, whether that support is a single Γ -orbit, the values $\lambda_k(u)$ at the generators with their minimal polynomials and orders, and the conductor, parity, and fundamental discriminant of λ_k . It checks (3.24), (3.27), $\sigma_u\sigma_v = \sigma_{uv}$, and the admissible completions of Proposition 4.7; each generator has one completion. Every block acts by center scalars and every quadratic block agrees with its Kronecker symbol at every generator.

5. INFINITE FAMILIES AND A COMPARISON PANEL

5.1. Groups of prime exponent. A single many-to-one power map and its weighted reverse map already create a noncommutative factor.

Theorem 5.1. *Suppose $\exp(G) = p$ is prime and write $k = |\text{Conj}(G)|$. Then*

$$\mathcal{P}_{\text{ram}}(G) \cong \begin{cases} M_2(\mathbb{Q}), & k = 2, \\ M_2(\mathbb{Q}) \oplus \mathbb{Q}, & k > 2. \end{cases} \quad (5.1)$$

In the defining representation, the scalar factor in the second line has multiplicity $k - 2$.

Proof. Let $e = e_1 \in \mathbb{Q}^k$, let $u = (1, \dots, 1)^\top$, and let $c = (c_1, \dots, c_k)^\top$ be the class-size vector, so $c_1 = 1$ and $\sum_i c_i = |G|$. Since $x^p = 1$ for every $x \in G$, we have $\sigma_p(i) = 1$ for all i , so (2.6) gives $P_p = ue^\top$; and $P_p^* = D_G^{-1} P_p^\top D_G = D_G^{-1} eu^\top D_G = (|G|/c_1) e (c^\top/|G|)$, so

$$P_p = ue^\top, \quad P_p^* = ec^\top. \quad (5.2)$$

The decomposition. Put $W = \text{span}_{\mathbb{Q}}\{e, u\}$ and

$$U = \{z \in \mathbb{Q}^k : z_1 = 0, c^\top z = 0\}.$$

Since $\langle z, e \rangle_G = z_1/|G|$ and $\langle z, u \rangle_G = c^\top z/|G|$, the space U is the class-orthogonal complement of W . If $k \geq 2$ then $u \notin \mathbb{Q}e$, so $\dim W = 2$; the two functionals $z \mapsto z_1$ and $z \mapsto c^\top z$ are linearly independent because $c \neq c_1 e = e$ when $k \geq 2$, so $\dim U = k - 2$. Also $W \cap U = 0$: if $\alpha e + \beta u \in U$ then its first coordinate gives $\alpha + \beta = 0$, while $c^\top(\alpha e + \beta u) = \alpha + \beta|G| = 0$; subtracting and using $|G| > 1$ forces $\alpha = \beta = 0$. Hence $\mathbb{Q}^k = W \oplus U$ orthogonally.

Both generators vanish on U : for $z \in U$, $P_p z = u(e^\top z) = z_1 u = 0$ and $P_p^* z = e(c^\top z) = 0$. Both preserve W , since $P_p e = u$, $P_p u = u$, $P_p^* e = c_1 e = e$ and $P_p^* u = (c^\top u) e = |G| e$. So W and U are reducing for P_p and P_p^* .

The action on W . In the ordered basis (e, u) the four operators $P_p, P_p^*, P_p P_p^*, P_p^* P_p$ restrict to

$$\begin{aligned} P_p|_W &= \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, & P_p^*|_W &= \begin{pmatrix} 1 & |G| \\ 0 & 0 \end{pmatrix}, \\ (P_p P_p^*)|_W &= \begin{pmatrix} 0 & 0 \\ 1 & |G| \end{pmatrix}, & (P_p^* P_p)|_W &= \begin{pmatrix} |G| & |G| \\ 0 & 0 \end{pmatrix}. \end{aligned} \quad (5.3)$$

Reading these as vectors in $\mathbb{Q}^4$ with coordinates $(a_{11}, a_{12}, a_{21}, a_{22})$ gives the matrix

$$\begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & |G| & 0 & 0 \\ 0 & 0 & 1 & |G| \\ |G| & |G| & 0 & 0 \end{pmatrix},$$

which is block anti-diagonal and has determinant $\pm|G|(|G| - 1)^2 \neq 0$ because $|G| > 1$. Hence the four restrictions are linearly independent and span $M_2(\mathbb{Q})$.

Conclusion. Every word in P_p, P_p^* of length at least one vanishes on U , by the previous paragraph, and the span of such words restricted to W is all of $M_2(\mathbb{Q})$ by the determinant computation. Therefore

$$\mathcal{P}_E(G) = \{a \oplus \lambda I_U : a \in \text{End}_{\mathbb{Q}}(W), \lambda \in \mathbb{Q}\},$$

the summand λI_U arising only from the adjoined identity. If $U = 0$, that is $k = 2$, this is $M_2(\mathbb{Q})$. If $U \neq 0$, the element $I_W \oplus 0$ lies in the algebra, so the algebra is $M_2(\mathbb{Q}) \oplus \mathbb{Q}$, and the scalar factor acts on U , which has dimension $k - 2$. $\square$

The theorem gives a family independent of the classification of the underlying groups. It applies to every elementary abelian p -group and, for odd p , to every extraspecial group of exponent p . All such groups with more than two classes have abstract algebra $M_2(\mathbb{Q}) \oplus \mathbb{Q}$, while the multiplicity $k(G) - 2$ of the scalar factor retains the class count at represented level.

5.2. Cyclic groups of prime-power order. There is a complete formula for cyclic prime powers.

Lemma 5.2. *Let F be the linear map on $\mathbb{Q}^{C_{p^a}}$ defined on coordinate vectors by $F\delta_x = \delta_{px}$. There is an orthogonal rational decomposition into reducing subspaces for F, F^* consisting of one chain of dimension $a + 1$, together with chains of dimension j having multiplicity*

$$m_a = p - 2, \quad m_j = p^{a-j-1}(p - 1)^2 \quad (1 \leq j < a). \quad (C)$$

Terms of multiplicity zero are absent. On a chain of dimension j , the algebra generated by F, F^ is $M_j(\mathbb{Q})$, and two chains of the same length are equivalent as modules over it.*

Proof. Write $C_{p^a} = \mathbb{Z}/p^a$ additively and let $\ell(x) = a - v_p(x)$ be the number of steps from x to 0 under $x \mapsto px$, so that $\ell(x) = j$ if and only if x has order p^j . There are $p^{j-1}(p - 1)$ vertices at level $j \geq 1$ and one at level 0. Every vertex at level $j \leq a - 1$ has p predecessors, all at level $j + 1$; the predecessors of 0 are 0 itself together with the $p - 1$ elements of order p . Vertices at level a are the leaves, and F^* annihilates them.

Note first that $F^*\delta_x = \sum_{py=x} \delta_y$, whence

$$FF^*\delta_x = \sum_{py=x} \delta_{py} = p\delta_x \quad (\ell(x) \leq a - 1), \quad (5.4)$$

and $FF^*\delta_x = 0$ on the leaves. Thus $FF^* = p \cdot I$ off the leaves. This is the source of the uniform chain weights below.

Nonradial chains. Fix a vertex v with $1 \leq \ell(v) = j \leq a - 1$ and let $w_0 = \sum_y c_y \delta_y$ run over the $(p - 1)$ -dimensional space of sibling differences on the predecessor set of v , so $\sum_y c_y = 0$. Put $w_k = (F^*)^k w_0$ and set $i = a - j$. Then $Fw_0 = (\sum_y c_y) \delta_v = 0$, while (5.4) gives

$$Fw_k = FF^*w_{k-1} = p w_{k-1} \quad (1 \leq k \leq i - 1), \quad (5.5)$$

because w_{k-1} is supported at level $j+k \leq a-1$. The vector w_k is supported at level $j+1+k$, so the chain terminates at w_{i-1} , and $F^*w_{i-1} = 0$ since that vector is supported on leaves. Vectors supported at distinct levels are orthogonal for the uniform form, so the w_k are an orthogonal rational basis of a reducing subspace of dimension i . Counting: level $j = a-i$ carries $p^{a-i-1}(p-1)$ vertices for $j \geq 1$, each contributing $p-1$ independent differences, which gives the multiplicity $p^{a-i-1}(p-1)^2$ of (C).

Distinct chains are orthogonal. This does not follow from level bookkeeping alone, because two chains born at different levels do have vectors at a common level, and their supports can even be nested. We verify it directly. Since $(F^*)^k \delta_y = \sum_{z \in F^{-k}(y)} \delta_z$, the chain vector born at v is

$$w_k^{(v)} = \sum_{y \in F^{-1}(v)} c_y \mathbf{1}_{F^{-k}(y)}, \quad \sum_y c_y = 0. \quad (5.6)$$

Let v, v' be birth vertices at levels $j \geq j'$ and let $w_k^{(v)}, w_{k'}^{(v')}$ both be supported at level L , so $k = L-j-1$ and $k' = L-j'-1$, whence $k' \geq k$. If $j = j'$ and $v \neq v'$ the fibres $F^{-1}(v)$ and $F^{-1}(v')$ are disjoint, hence so are the sets $F^{-k}(y)$ occurring in (5.6), and the inner product vanishes. If $j > j'$, then $k' > k$, and for $y \in F^{-1}(v)$, $y' \in F^{-1}(v')$ the intersection $F^{-k}(y) \cap F^{-k'}(y')$ is nonempty only if $y' = F^{k'-k}(y)$, in which case $F^{-k}(y) \subseteq F^{-k'}(y')$ and the intersection has p^k elements. But every $y \in F^{-1}(v)$ satisfies $F(y) = v$, so for $k' - k \geq 1$ the value $F^{k'-k}(y) = F^{k'-k-1}(v) =: y'_0$ is the same vertex for all such y . Therefore

$$\langle w_k^{(v)}, w_{k'}^{(v')} \rangle = p^k c'_{y'_0} \sum_{y \in F^{-1}(v)} c_y = 0.$$

The same computation against the radial vectors $r_L = \mathbf{1}_{\ell^{-1}(L)}$ gives $\langle w_k^{(v)}, r_L \rangle = p^k \sum_y c_y = 0$. Hence all the chains listed here and below are mutually orthogonal.

The radial chain. Put $r_k = \sum_{\ell(x)=k} \delta_x$ for $0 \leq k \leq a$. Then

$$\begin{aligned} Fr_0 &= r_0, & Fr_1 &= (p-1)r_0, & Fr_k &= pr_{k-1} \quad (k \geq 2), \\ F^*r_0 &= r_0 + r_1, & F^*r_k &= r_{k+1} \quad (1 \leq k \leq a-1), & F^*r_a &= 0. \end{aligned} \quad (5.7)$$

These $a+1$ orthogonal vectors span a reducing subspace. The predecessor set of 0 contains 0, so F has the eigenvector r_0 and is not nilpotent here; this is the one chain on which the endpoint argument for a nilpotent shift is unavailable.

Chains born at the origin. The vertex 0 is the one birth vertex whose predecessor set is not contained in a single level: $F^{-1}(0)$ consists of 0 itself together with the $p-1$ elements of order p . Among the $(p-1)$ -dimensional space of sibling differences on $F^{-1}(0)$, those not involving δ_0 , that is, supported on the level-one vertices alone, form a subspace of dimension $(p-1)-1 = p-2$. For such a w_0 we again have $Fw_0 = (\sum_y c_y)\delta_0 = 0$, and w_0 is supported at level one, so the chain $w_k = (F^*)^k w_0$ terminates at w_{a-1} and has dimension a . This gives the multiplicity $m_a = p-2$, absent when $p=2$. The one remaining direction on $F^{-1}(0)$, the one involving δ_0 , is already accounted for by the radial chain, as the dimension count in Theorem 5.3 confirms. Note that $i=a$ does not arise from the nonradial analysis above, which required $1 \leq j \leq a-1$, so the length- a chains are these $p-2$ chains.

The algebra restricted to each chain is a full matrix algebra. Let W be any of the above chains, with ordered spanning vectors $w_0, \dots, w_{i-1}$ in the nonradial case or $r_0, \dots, r_a$ in the radial case, and suppose $S \in \text{End}_{\mathbb{Q}}(W)$ commutes with $F|_W$ and $F^*|_W$. By (5.5) and (5.7), $F^*|_W$ has one-dimensional kernel, spanned by the top vector w_{i-1} , respectively r_a . Hence S preserves that line and acts on it by some $\lambda \in \mathbb{Q}$. Applying F repeatedly and using that every weight in (5.5) and (5.7) is nonzero, the weights being p , and $p-1 \geq 1$ at the last radial step, gives $S = \lambda$ on each successive vector in turn. Therefore $(\mathcal{P}_E|_W)' = \mathbb{Q}$, and Theorem 2.14 gives

$$\mathcal{P}_E|_W = (\mathcal{P}_E|_W)'' = \text{End}_{\mathbb{Q}}(W) \cong M_{\dim W}(\mathbb{Q}).$$

This argument is uniform in the two cases and does not use nilpotency.

Finally, two chains of equal length are isomorphic as modules. For the nonradial chains, including those born at the origin, (5.5) shows that F acts as p times the downward shift and F^* as the upward shift, with the same weight p at every step, independently of length or birth vertex; so any two such chains of the same length are carried to one another by the linear map matching their spanning vectors in order. The radial chain is not of this form, since its last step has weight $p - 1$ and F is not nilpotent on it, but it has dimension $a + 1$, whereas every nonradial chain has dimension at most a , so no comparison between the two types is required. $\square$

Theorem 5.3. *Let $a \geq 1$. Then*

$$\mathcal{P}_{\text{ram}}(C_{p^a}) \cong M_{a+1}(\mathbb{Q}) \oplus \bigoplus_{j=1}^a M_j(\mathbb{Q}), \quad p \text{ odd}, \quad (5.8)$$

$$\mathcal{P}_{\text{ram}}(C_{2^a}) \cong M_{a+1}(\mathbb{Q}) \oplus \bigoplus_{j=1}^{a-1} M_j(\mathbb{Q}). \quad (5.9)$$

Empty sums are omitted. The M_{a+1} -module occurs once. For odd p , the multiplicity of the M_a -module is $p - 2$, and for $1 \leq j < a$ the multiplicity of the M_j -module is

$$m_j = p^{a-j-1}(p-1)^2.$$

For $p = 2$, the M_a -block is absent and the same formula gives $m_j = 2^{a-j-1}$ for $j < a$.

Proof. Write C_{p^a} additively. Its class measure is uniform, so the sole canonical generator is pullback P along $x \mapsto px$, together with the ordinary transpose $P^\top$. The transpose is the map F of Lemma 5.2. That lemma gives the displayed simple factors and their multiplicities. The dimension identity

$$p^a = (a+1) + a(p-2) + \sum_{j=1}^{a-1} j p^{a-j-1}(p-1)^2$$

shows that the listed chains exhaust the coordinate space. $\square$

Corollary 5.4. *For odd p ,*

$$\dim_{\mathbb{Q}} \mathcal{P}_{\text{ram}}(C_{p^a}) = (a+1)^2 + \sum_{j=1}^a j^2,$$

whereas for $p = 2$ the final summand a^2 is omitted. These dimensions depend only on a and on whether $p = 2$, while the represented multiplicities retain p .

5.3. Dihedral and quaternion groups. Use the presentations

$$D_{2n} = \langle a, b : a^n = b^2 = 1, bab^{-1} = a^{-1} \rangle, \quad Q_{4m} = \langle a, b : a^{2m} = 1, b^2 = a^m, bab^{-1} = a^{-1} \rangle,$$

with all prime-power maps dividing the exponent. Modular row reduction followed by rational reconstruction gives the following table.

Proposition 5.5. *For $3 \leq n \leq 16$ and $2 \leq m \leq 8$, the represented algebras $A = \mathcal{P}_{\text{ram}}(G)$ have the signatures*

G	$ G $	k	$\dim A$	$\dim Z$	$\dim A'$	G	$ G $	k	$\dim A$	$\dim Z$	$\dim A'$
D_6	6	3	9	1	1	D_{22}	22	7	13	5	5
D_8	8	5	17	2	2	D_{24}	24	9	51	3	3
D_{10}	10	4	10	2	2	D_{26}	26	8	14	6	6
D_{12}	12	6	26	2	2	D_{28}	28	10	34	4	4
D_{14}	14	5	11	3	3	D_{30}	30	9	31	4	4
D_{16}	16	7	26	2	5	D_{32}	32	11	41	3	11
D_{18}	18	6	18	3	3						
Q_8	8	5	10	2	5	Q_{24}	24	9	51	3	3
Q_{12}	12	6	26	2	2	Q_{28}	28	10	34	4	4
Q_{16}	16	7	26	2	5	Q_{32}	32	11	41	3	11
Q_{20}	20	8	30	3	3						

Multiplicity freeness holds at every rung except D_{16} , D_{32} , Q_8 , Q_{16} , and Q_{32} , where $\dim A' > \dim Z$.

Proof. Rational computation as described above; the multiplicity statement is Corollary 2.21 applied to the commutant and center columns. $\square$

Theorem 5.6. *Let p be an odd prime, put $m = (p - 1)/2$, and let*

$$d = \text{ord}_{U_p/\{\pm 1\}}(2), \quad q = m/d.$$

Then

$$\mathcal{P}_{\text{ram}}(D_{2p}) \otimes_{\mathbb{Q}} \mathbb{C} \cong M_3(\mathbb{C}) \oplus \mathbb{C}^{d-1+\epsilon_q}, \quad \epsilon_q = \begin{cases} 0, & q = 1, \\ 1, & q > 1. \end{cases}$$

Hence

$$\dim_{\mathbb{Q}} \mathcal{P}_{\text{ram}}(D_{2p}) = 8 + d + \epsilon_q.$$

The class-coordinate representation is multiplicity free if and only if $q = 1$.

Proof. Let R be the span of the m rotation-class coordinates and let $\mathbf{1}_R$ be their sum. The space

$$W = \text{span}_{\mathbb{C}}\{\delta_1, \delta_s, \mathbf{1}_R\}$$

reduces Ψ_2, Ψ_p and their class-form adjoints. Their restrictions to W , computed with class weights $1, p, p - 1$, generate $M_3(\mathbb{C})$.

On $W^\perp$, the operators Ψ_p and Ψ_p^* vanish, while Ψ_2 is the permutation induced by multiplication by 2 on $(\mathbb{Z}/p\mathbb{Z})^\times/\{\pm 1\}$. This permutation has q cycles of length d . Its eigenvalues are the d -th roots of unity. Removing $\mathbf{1}_R$ removes the eigenvalue 1 when $q = 1$; for $q > 1$ its multiplicity becomes $q - 1$. Thus the algebra on $W^\perp$ is $\mathbb{C}^{d-1+\epsilon_q}$. The commutator ideal supplies the $M_3(\mathbb{C})$ summand, so the two restrictions form the displayed direct sum.

For $q > 1$, each nontrivial d -th root occurs with multiplicity q ; for $q = 1$, all eigenspaces on $W^\perp$ are one-dimensional. This gives the last assertion. $\square$

For $p = 17$, $d = 4$ and $q = 2$, so

$$\mathcal{P}_{\text{ram}}(D_{34}) \otimes_{\mathbb{Q}} \mathbb{C} \cong M_3(\mathbb{C}) \oplus \mathbb{C}^4, \quad \dim_{\mathbb{Q}} \mathcal{P}_{\text{ram}}(D_{34}) = 13.$$

5.4. Rational classes in $\text{PSL}(2, q)$.

Theorem 5.7. *Let $q = p^f$, let $G = \text{PSL}(2, q)$, and put*

$$m_{\pm} = \frac{q \pm 1}{\gcd(2, q - 1)}.$$

- (1) Let $x \in G$ have order d with $p \nmid d$. Then d divides m_+ or m_- , and the rational class of $[x]$ has $\varphi(d)/2$ elements if $d > 2$ and one element if $d \leq 2$.
- (2) The rational class of a class of p -elements has at most two elements.
- (3) Let r be an odd prime. Then G has a rational class of size r if and only if $\varphi(d) = 2r$ for some $d > 2$ dividing m_+ or m_- . Such a q exists if and only if $2r + 1$ is prime.

Proof. (1) A nontrivial semisimple element of G lies in a maximal torus, and the maximal tori of G are cyclic of order m_+ or m_- ; this gives the divisibility. Let $d > 2$, let T be a maximal torus containing x , and let u be a good label, so that $x^u \in T$ also has order d . For $d > 2$ we have $C_G(x) = C_G(x^u) = T$, so if $gx^ug^{-1} = x$ then $gTg^{-1} = T$, that is, $g \in N_G(T)$. Since $N_G(T)/T$ has order two and acts on T by inversion, x^u is conjugate to x if and only if $u \equiv \pm 1 \pmod{d}$. Thus the stabilizer of $[x]$ in U_N is the preimage of $\{\pm 1\} \subseteq (\mathbb{Z}/d)^\times$, of index $\varphi(d)/2$. For $d \leq 2$ every good label acts trivially on $\langle x \rangle$.

(2) If q is even the p -elements are the involutions and (1) applies. If q is odd, every nontrivial unipotent element of $\mathrm{SL}(2, q)$ has order p and there are two unipotent classes, so G has two classes of p -elements in all.

(3) By (1) and (2) the sizes of the rational classes are the numbers $\varphi(d)/2$ for $d > 2$ dividing m_+ or m_- , together with values at most two, and $r \geq 3$; this is the stated criterion. If $\varphi(d) = 2r$ then d is ℓ or 2ℓ with $\ell = 2r + 1$ prime, except for $r = 3$, where also $d \in \{9, 18\}$; in every case $2r + 1$ is prime. Conversely if $\ell = 2r + 1$ is prime, choose by Dirichlet a prime $q \equiv 1 \pmod{2\ell}$; then ℓ divides $m_- = (q - 1)/2$ and $\varphi(\ell) = 2r$. $\square$

Corollary 5.8. *No $\mathrm{PSL}(2, q)$ has a rational class of size 7.*

Proof. By Theorem 5.7(3) it suffices that 14 is not a value of φ . If $\varphi(d) = 14$ then $\varphi(p^a) \mid 14$ for every prime power $p^a \parallel d$, so $\varphi(p^a) \in \{1, 2, 14\}$. For odd p , $\varphi(p^a) = 2$ gives $p^a = 3$, while $\varphi(p^a) = 14$ forces $p \mid 14$, and $\varphi(7) = 6$, $\varphi(49) = 42$. Hence every factor contributes a power of two and $\varphi(d)$ is a power of two. $\square$

For $3 \leq q \leq 32$ the sizes $\varphi(d)/2$ give odd prime values only for

$$\begin{aligned} \mathrm{PSL}(2, 8) : d = 7, 9, r = 3, & \quad \mathrm{PSL}(2, 13) : d = 7, r = 3, & \quad \mathrm{PSL}(2, 17) : d = 9, r = 3, \\ \mathrm{PSL}(2, 19) : d = 9, r = 3, & \quad \mathrm{PSL}(2, 27) : d = 7, 14, r = 3, & \quad \mathrm{PSL}(2, 23) : d = 11, r = 5, \\ \mathrm{PSL}(2, 29) : d = 7, 14, r = 3, & \quad \mathrm{PSL}(2, 32) : d = 11, r = 5, \end{aligned}$$

and for no other q in that range; in particular $\mathrm{PSL}(2, 7)$, $\mathrm{PSL}(2, 9)$, $\mathrm{PSL}(2, 11)$, $\mathrm{PSL}(2, 16)$, $\mathrm{PSL}(2, 25)$ have none.

5.5. A comparison panel.

Theorem 5.9. *For each of the following 31 groups the rational algebra $A = \mathcal{P}_{\mathrm{ram}}(G)$ has the stated dimension, center dimension z , and represented commutant dimension c ; the pairs (d, m) give the complex blocks $M_d(\mathbb{C})$ and their multiplicities in $V_G \otimes_{\mathbb{Q}} \mathbb{C}$.*

G	$ G $	$k(G)$	$\exp G$	$\dim A$	z	c	(d, m)
<i>Cyclic</i>							
C_2	2	2	2	4	1	1	(2, 1)
C_3	3	3	3	5	2	2	(2, 1), (1, 1)
C_4	4	4	4	10	2	2	(3, 1), (1, 1)
C_5	5	5	5	5	2	10	(2, 1), (1, 3)
C_6	6	6	6	20	2	2	(4, 1), (2, 1)
C_8	8	8	8	21	3	6	(4, 1), (2, 1), (1, 2)
C_{12}	12	12	12	50	4	4	(6, 1), (3, 1), (2, 1), (1, 1)
<i>Elementary abelian</i>							
$C_2 \times C_2$	4	4	2	5	2	5	(2, 1), (1, 2)

G	$ G $	$k(G)$	$\exp G$	$\dim A$	z	c	(d, m)
C_2^3	8	8	2	5	2	37	$(2, 1), (1, 6)$
$C_3 \times C_3$	9	9	3	5	2	50	$(2, 1), (1, 7)$
<i>Dihedral</i>							
D_8	8	5	4	17	2	2	$(4, 1), (1, 1)$
D_{10}	10	4	10	10	2	2	$(3, 1), (1, 1)$
D_{12}	12	6	6	26	2	2	$(5, 1), (1, 1)$
D_{16}	16	7	8	26	2	5	$(5, 1), (1, 2)$
<i>Quaternion</i>							
Q_8	8	5	4	10	2	5	$(3, 1), (1, 2)$
<i>Symmetric</i>							
S_3	6	3	6	9	1	1	$(3, 1)$
S_4	24	5	12	25	1	1	$(5, 1)$
S_5	120	7	60	49	1	1	$(7, 1)$
S_6	720	11	60	59	3	3	$(7, 1), (3, 1), (1, 1)$
S_7	5040	15	420	225	1	1	$(15, 1)$
S_8	40320	22	840	442	2	2	$(21, 1), (1, 1)$
<i>Alternating</i>							
A_4	12	4	6	10	2	2	$(3, 1), (1, 1)$
A_5	60	5	30	17	2	2	$(4, 1), (1, 1)$
A_6	360	7	60	27	3	3	$(5, 1), (1, 1), (1, 1)$
A_7	2520	9	420	65	2	2	$(8, 1), (1, 1)$
A_8	20160	14	420	146	3	3	$(12, 1), (1, 1), (1, 1)$
<i>Simple of Lie type</i>							
$\text{PSL}(2, 7)$	168	6	84	26	2	2	$(5, 1), (1, 1)$
$\text{PSL}(2, 8)$	504	9	126	29	5	5	$(5, 1), (1, 1), (1, 1), (1, 1), (1, 1)$
$\text{PSL}(2, 11)$	660	8	330	38	3	3	$(6, 1), (1, 1), (1, 1)$
<i>Nonabelian of order 27</i>							
H_{27}	27	11	3	5	2	82	$(2, 1), (1, 9)$
<i>Frobenius of order 20</i>							
F_{20}	20	5	20	17	2	2	$(4, 1), (1, 1)$

Proof. Rational row reduction gives the dimensions of the algebra, center, and commutant, as in Section 6. It also enumerates the positive integral solutions of

$$\sum_j d_j^2 = \dim A, \quad \sum_j m_j^2 = \dim A', \quad \sum_j d_j m_j = k(G), \quad \#\{j\} = \dim Z(A), \quad (5.10)$$

and for every row the listed solution is unique. Modular receipts at 1,000,003 and 1,000,033 reproduce every dimension. $\square$

The groups D_8 and Q_8 both have five classes and identical character tables, with $\dim A = 17$ and $\dim A = 10$; the invariant is therefore strictly finer than the character table. Among the symmetric groups S_3, S_4, S_5 , and S_7 give full matrix algebras, whereas S_6 and S_8 do not. The scalar block of Q_8 occurs twice in class-coordinate space, which is where its commutant differs from that of D_8 . Both H_{27} and $C_3 \times C_3$ have $\dim A = 5$, with $c = 82$ and $c = 50$.

6. COMPUTATIONAL VERIFICATION

The computations use character-table data rather than matrix or permutation realizations. GAP [10] and CTblLib [4] supply the ordered conjugacy classes, class sizes, exponents, and prime-power maps; GAP stores an n th power map as a list whose i th entry gives the class position of the n th powers of the elements of class i [12]. For a table with k classes the runner builds P_n and

$D^{-1}P_n^\top D$ over $\mathbb{Q}$, adjoins products until multiplication by every generator adds no new rational row, and obtains the center and represented commutant from rational commutator nullspaces; a separating central element and its factorization give the central idempotents, whose ideal dimensions give the split degrees over the center fields. FLINT performs the rational rank calculations [9].

Reductions modulo 1,000,003 and 1,000,033 repeat every dimension calculation. The proof of each stated decomposition is the rational word basis together with the central factorization; the modular runs are consistency checks. The GAP exports, source, tests, and JSON outputs accompany this paper as ancillary files, together with a supplement recording the stopping criteria, schema, file hashes, and reproduction commands.

The principal block of each sporadic algebra is certified as follows. Since every element of G has order dividing $N = \exp(G)$, the operator $|G|^{-1}\Psi_N^*\Psi_N$ is the coordinate projection onto the identity class, so $E_{1A} \in A$. The cyclic module $W = A\delta_{1A}$ is computed by exact rational orbit closure, and Lemma 4.18 gives $A|_W = \text{End}_{\mathbb{Q}}(W)$; the degree of the principal block is therefore $\dim_{\mathbb{Q}} W$, replayed at two primes by nonzero orbit minors. The residual algebra on the orthogonal complement of W is then closed and decomposed separately. This replaces an earlier procedure that took the number of cells of the stable weighted refinement as the principal degree. That partition is equitable, but equitability does not imply fullness, and for Fi_{23} it overcounts: the refinement has 92 cells while $\dim_{\mathbb{Q}} A\delta_{1A} = 91$, the missing dimension being the invariant line of Section 4.5. The other twenty-five sporadic rows are unchanged by the corrected procedure.

Independently of the word-basis ranks, every reported row is checked against the represented Wedderburn identities

$$\sum_j d_j^2 = \dim A, \quad \sum_j m_j^2 = \dim A', \quad \sum_j d_j m_j = k(G), \quad \sum_j [K_j : \mathbb{Q}] = \dim_{\mathbb{Q}} Z(A),$$

where conjugate complex blocks are grouped over their rational center. These checks are especially restrictive in the largest rows. For the Monster they give

$$170 + 5 + 3 + 4 \cdot 2 + 8 = 194, \quad 170^2 + 5^2 + 3^2 + 4 \cdot 2^2 + 8 = 28,958,$$

with fifteen multiplicity-one rational blocks. For $O'N$, the quadratic center contributes two dimensions to the center and the class count forces one scalar block of multiplicity two. For $\mathbb{B}$, the center and commutant dimensions force two multiplicity-two blocks. Thus the tables are subject both to exact algebra closure and to the full set of representation identities, not only to modular rank agreement.

The ramified local factors are computed by two routes. The first forms the $I_{p,m}$ -orbits and the induced Frobenius permutation; the second uses, for each rational-class stabilizer $S \leq U_m$,

$$f = \text{ord}_{U_m/SI_{p,m}}(\kappa_{p,m}), \quad g = [U_m : SI_{p,m}]/f,$$

and the factor $(1 - T^f)^g$. The two routes agree in all 173 sporadic bad-prime slots. The exported ramified maps generate the required U_m -action in 168 slots; the other five are omitted from the computed table. Cyclic groups through order 72 and eight abelian groups give the cyclotomic local factors independently.

7. FURTHER PROBLEMS

The inverse problem asks which semisimple $\mathbb{Q}$ -algebras occur as $\mathcal{P}_{\text{ram}}(G)$, or as $\mathcal{P}_E(\mathcal{X})$ with the class-coordinate module attached. The identities (5.10) are necessary but far from sufficient: by Theorem 5.1 every group of prime exponent with $k > 2$ gives $M_2(\mathbb{Q}) \oplus \mathbb{Q}$ whatever its order. For the extreme case Corollary 2.20 converts the question into rigidity: $\dim A = k^2$ if and only if the weighted colored power graph has no balanced kernel beyond the scalars.

By Theorem 5.7 the odd primes r for which $2r + 1$ is composite are unrealized in $\text{PSL}(2, q)$, the first being $r = 7$. They are realized by abelian groups: C_{29} has a single nonidentity rational class,

of size 28, and hence a summand with endomorphism field $\mathbb{Q}(\zeta_7)$. Which $\mathbb{Q}(\zeta_r)$ occur as centers of blocks of $\mathcal{P}_{\text{ram}}(G)$ for G simple is open; the value $r = 7$ occurs for no sporadic group.

The Monster, Co_2 , Co_3 , and Fi_{23} each have a nonprincipal simple factor inside the Galois-fixed part of V_G (Section 4.5). For the Monster it comes from a permutation of classes; for Fi_{23} it does not. Which finite groups admit symmetries of the weighted power system that are not automorphisms of the character table, and which admit balanced kernels beyond the span of all such symmetries, are open.

Equality in the synchronized tensor containment (2.26) is open.

CODE AND DATA AVAILABILITY

The GAP and CTblLib exports, the rational word-closure runner, the Monster certificate, and the JSON outputs underlying Theorem 5.9, Theorem 4.2, Theorem 4.8, and Theorem 4.22 are included as ancillary files, with a single reproduction script and a test suite. Section 6 describes the checks they perform.

APPENDIX A. POWER DYNAMICS AND THE SPORADIC SEPARATION THEOREM

The forward operators retain a finite dynamical invariant that is different from the adjoint-closed algebra. Let $\tau : X \rightarrow X$ be a map of a finite set and let P_τ act on $\mathbb{Q}^X$ by pullback, $(P_\tau f)(x) = f(\tau x)$. Write $c_\ell(\tau)$ for the number of directed cycles of length ℓ in the functional graph of τ .

Theorem A.1 (Finite-map zeta formula). *For every finite map $\tau : X \rightarrow X$,*

$$Z_\tau(z) := \det(I - zP_\tau)^{-1} = \prod_{\ell \geq 1} (1 - z^\ell)^{-c_\ell(\tau)}. \quad (\text{A.1})$$

Thus Z_τ depends only on the periodic cycles. The rooted trees feeding those cycles contribute no factor.

Proof. Order the vertices of each connected functional-graph component by decreasing distance from its unique directed cycle, followed by the cycle vertices. In this basis P_τ is block triangular. Its off-cycle diagonal block is nilpotent, while a cycle of length ℓ contributes the permutation matrix of an ℓ -cycle. Hence

$$\det(I - zP_\tau) = \prod_{\text{cycles } C} \det(I - zP_C) = \prod_{\text{cycles } C} (1 - z^{|C|}).$$

□

For a finite group G , put $N_G = \exp(G)$ and

$$\mathcal{N}_G = \{p^a : p \text{ prime}, a \geq 1, p^a \mid N_G\}.$$

Let $\sigma_n : \text{Conj}(G) \rightarrow \text{Conj}(G)$ be $C \mapsto C^n$. The *ramified dynamical packet* is the unlabelled multiset

$$\mathfrak{Z}_{\text{ram}}(G) = \{\{Z_{\sigma_n}(z) : n \in \mathcal{N}_G\}\}. \quad (\text{A.2})$$

It forgets both the prime-power labels and the ordering of the conjugacy classes. It is therefore weaker than the fully marked eidetic object.

Theorem A.2 (Sporadic dynamical separation). *For the twenty-six sporadic simple groups:*

- (1) *the map $G \mapsto \mathfrak{Z}_{\text{ram}}(G)$ is injective;*
- (2) *the cycle lengths occurring among all σ_n , $n \in \mathcal{N}_G$, lie in $\{1, 2, 3, 5\}$;*
- (3) *for each G , the set of occurring cycle lengths equals $\{1\}$ together with the orders of the good-label characters in Theorem 4.8;*

(4) a cycle of length 3 occurs exactly for

$$J_1, J_3, J_4, Ly, O'N, Ru,$$

and a cycle of length 5 occurs exactly for Ly .

Proof. The class power maps in the fixed CTblLib ordering give 258 prime-power directions. For each direction, cycle enumeration followed by Theorem A.1 gives its exponent vector (c_1, c_2, c_3, c_5) . Sorting these vectors within each group produces twenty-six distinct multisets, proving (1). Direct enumeration gives (2). Comparing the cycle-order support with the independently reconstructed good-label characters of Theorem 4.8 proves (3). The same exact enumeration gives the two lists in (4). $\square$

Corollary A.3 (Odd cyclotomic detection). *Among the sporadic simple groups, the ramified dynamical packet detects every nonrational center field in $\mathcal{P}_{\text{ram}}(G)$. Cubic cycles occur exactly for the six groups carrying a $\mathbb{Q}(\zeta_3)$ -factor, while the unique quintic cycle occurs for the unique group carrying a $\mathbb{Q}(\zeta_5)$ -factor.*

Proof. Combine Theorem A.2 with Theorems 4.2 and 4.8. $\square$

Remark A.4. The equality of supports in Theorem A.2(3) is a census theorem, not a general equivalence for finite groups. The general implication supplied by Lemma 3.18 is one-sided: character order is bounded by permutation order on block support.

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